Individual Assessment Coversheet

Campus: Tyger Valley

Faculty: Information Technology

Module Code: ITLSA1-22

Group: 4

Lecturer's Name: Ntombesisa Mateyisi

Student Full Name: Keenan Paiva Batista

Student Number: 4845350

Indicate	Yes	No
Plagiarism report attached		

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

Signature	Date
Bista	23 June 2024

Lecturer's Comments:

Marks Awarded:	%
Signature	**Date**

Table of Contents

Section A

Question 1

The IT solutions company, TechCorp, wishes to use an operating system between Ubuntu Linux and Kali Linux that is the most compatible with their requirements of web development, network security and penetration testing. Both distributors provide their versions of Linux as free and open-source, however both distributors offer their own unique pros and cons.

The history of Ubuntu Linux consists of Mark Shuttleworth with a group of open-source developers creating a new version of Linux incorporating a GNOME desktop environment and a Debian foundation. Ubuntu is the most widely used version of Linux for workstations globally and many special editions of Ubuntu were made for servers as well as OpenStack clouds. Ubuntu was also made to minimise professional services costs like maintenance and management (Ubuntu, 2024).

Kali Linux is a distributor that provides an open-source, Debian-based Linux operating system that was built off several previous developments. It was made for the purposes of acting as a solution to security professionals by specializing in Information Security activities like penetration testing. This can be done as Kali Linux provides many common tools, configurations as well as specifications (Kali Linux, 2024).

The benefits of Ubuntu Linux can be seen as Ubuntu Linux being open-source, it allows for its exceptional community of developers to develop and offer support such as updates and bug fixes. In comparison to the Windows operating system, Ubuntu Linux is less vulnerable to threats of security like malware. Furthermore, it also comes with a built-in firewall offering additional protection. Moreover, it is a very customizable operating system and allows users to add tools to perform specific actions (Nonis, 2024). Additionally, it provides a user-friendly interface akin to other operating systems like Windows and macOS (Sajee, 2023).There are also several free applications and software that can be installed on Ubuntu.

The detriments of Ubuntu Linux are seen through the limited number of applications and programs that can be used on this operating system. In addition, official technical support can be more limited than commercial operating systems despite community support. There can also be situations where latest or specialised hardware components can result in compatibility issues with Ubuntu.

The benefits of Kali Linux include the fact that Kali Linux receives regular updates and security updates developed by proficient groups ensuring that users have the latest features. Furthermore, this operating system is specifically designed for security professionals with strong security measures as well as configurations. It can also function as a platform for ethical hacking to take place on. It comes with pre-installed tools to accomplish several security activities such as penetration testing, analysis of a network and vulnerable assessment. Moreover, Kali Linux is flexible due to its environment allowing customization for users to use to meet their specific needs (Digital Djinn, 2023).

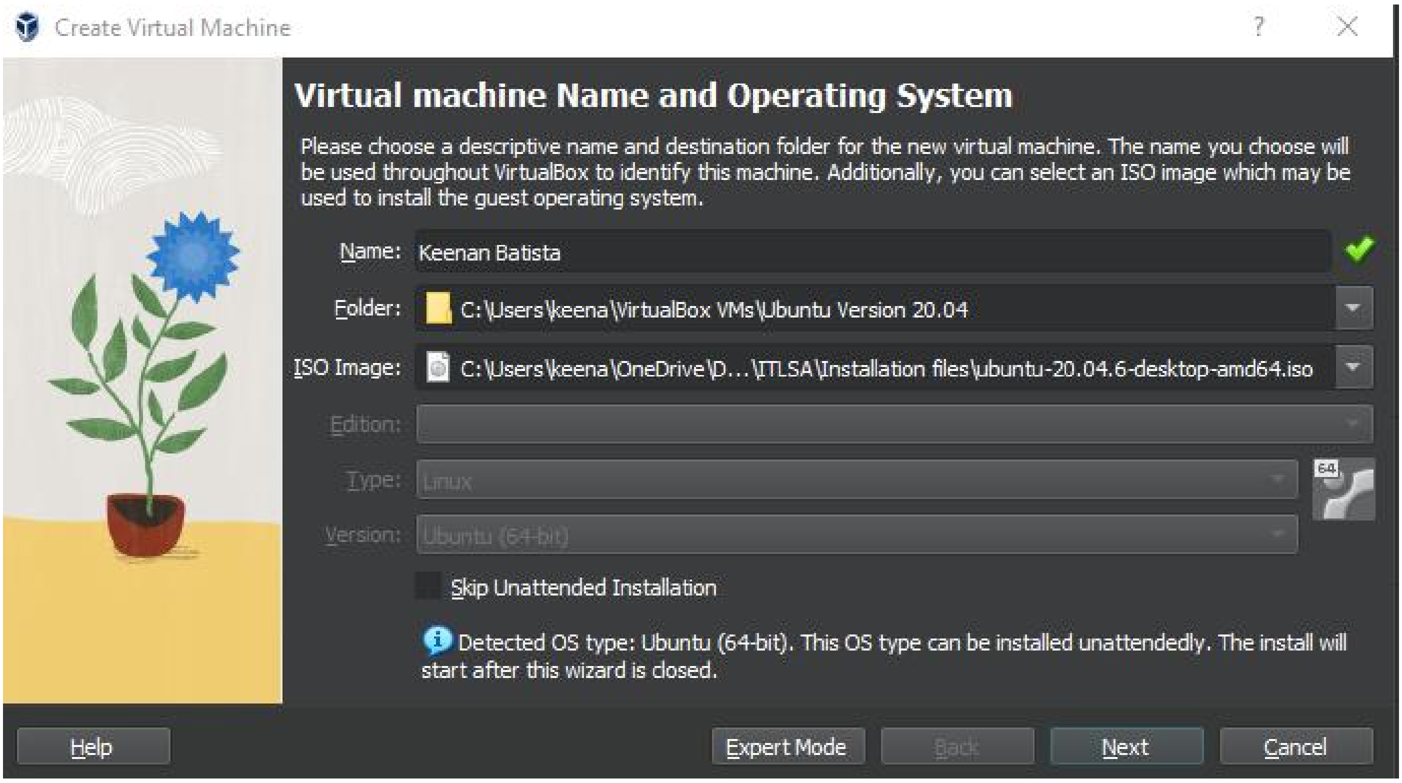
The detriments of Kali Linux consist of performance issues can occur due to the resource-incentive tools pre-installed on Kali Linux. Additionally, it can be used maliciously due to its relation to ethical hacking and testing for security. It is also not suitable for activities and tasks outside of security compared to other Linux software (Digital Djinn, 2023).

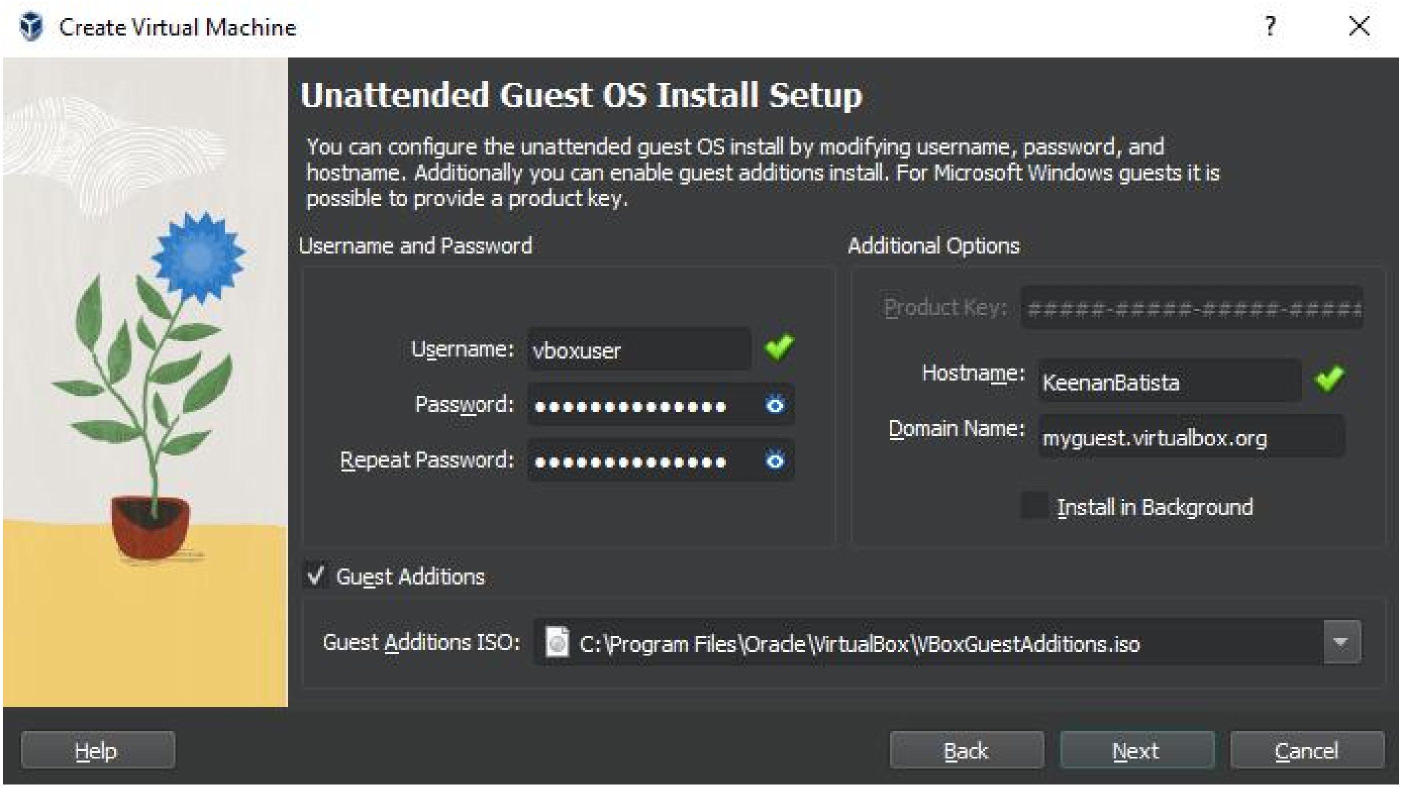
When evaluating the advantages and disadvantages of both Ubuntu and Kali Linux, Kali Linux is most suitable for TechCorp. Despite not having a user-friendly interface compared to the Ubuntu, Kali Linux provides all the necessary tools for all aspects of network security and perform penetration testing. Moreover, the Kali Linux environment can be customized to make it more suitable for web development with the tools that the operating system also delivers.

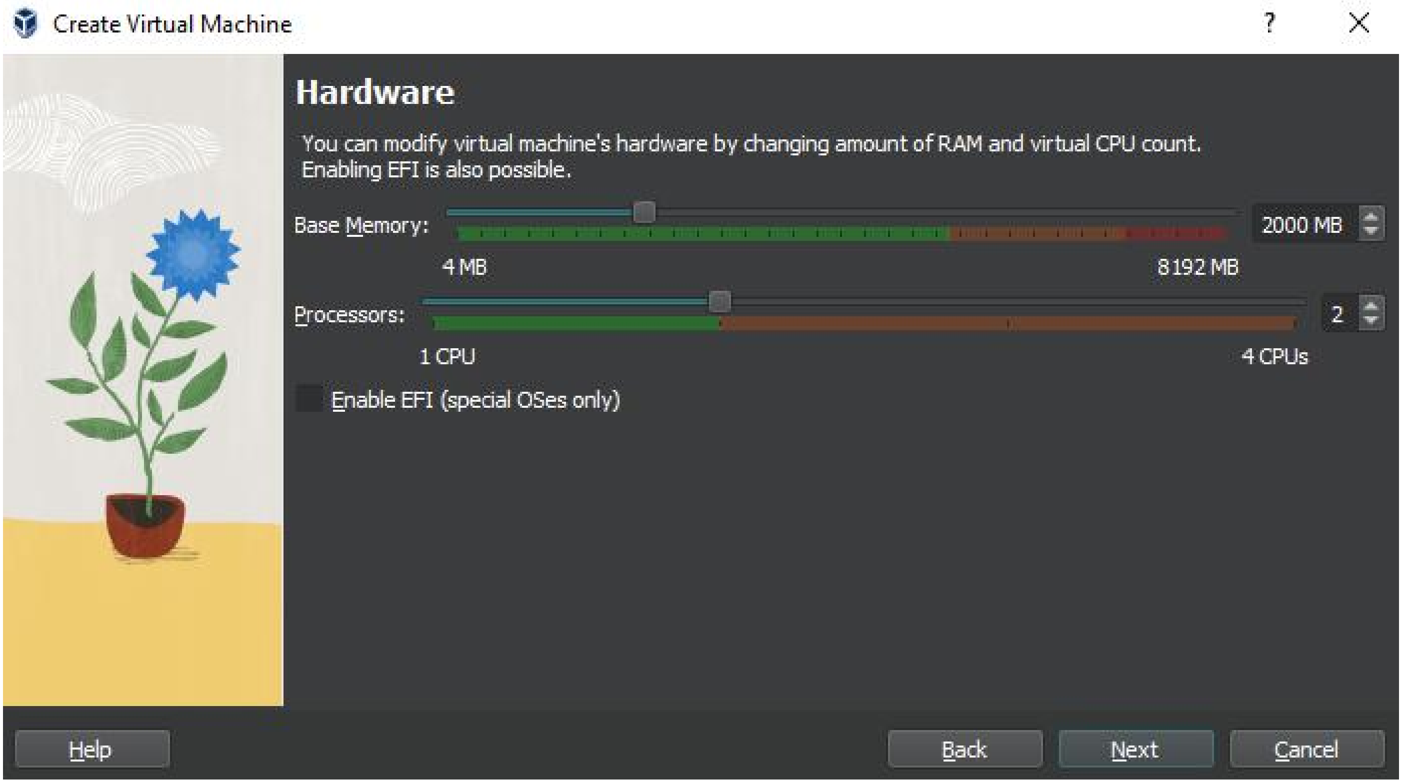
In conclusion, Khai Linux proves to be the most appropriate Linux distribution that can fufill the requirements needed by TechCorp.

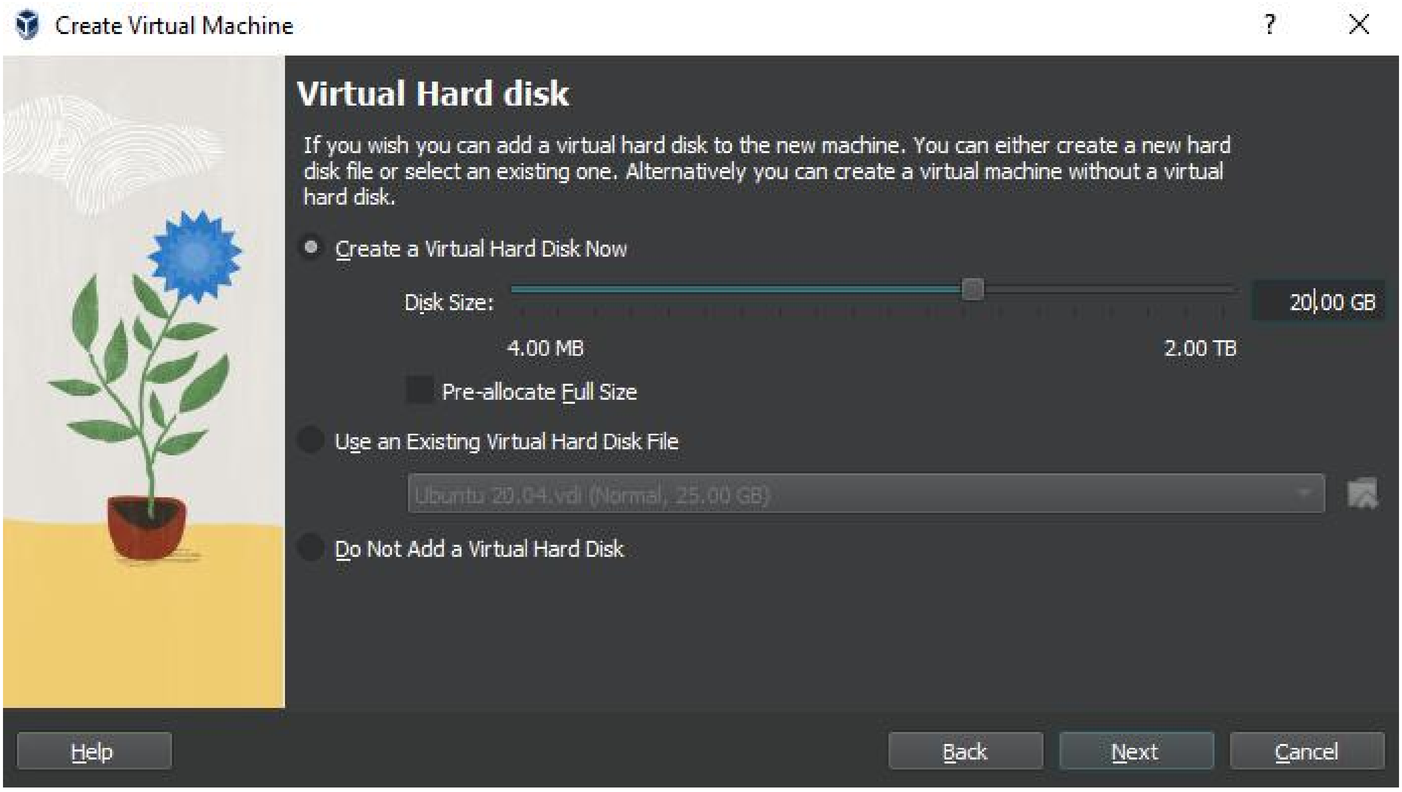
Question 2

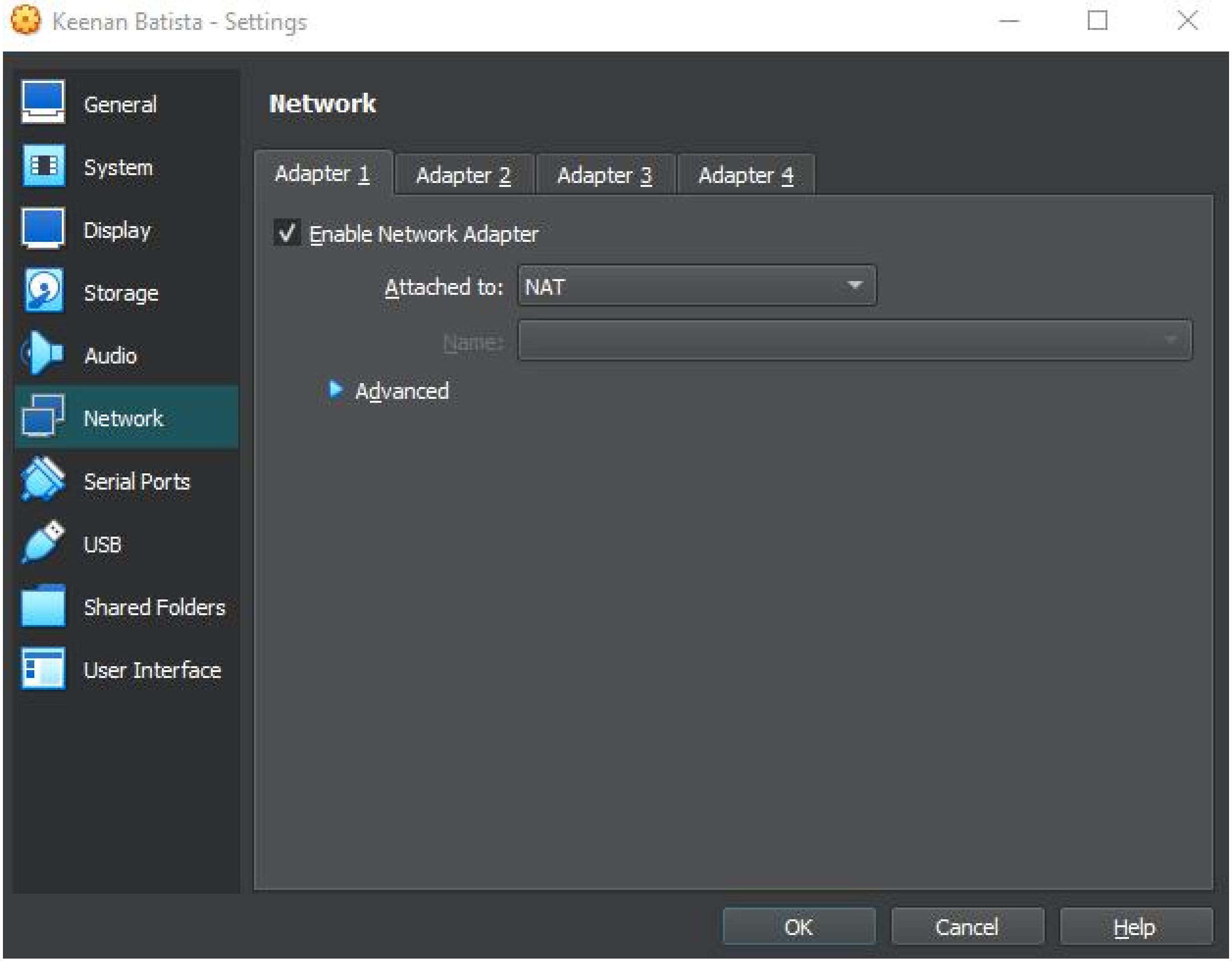
2.1)






Create Virtual Machine
Hardware
You can modify virtual machine's hardware by changing amount of RAM and virtual CPU count. Enabling EFI is also possible.
Base Memory:
2000 MB
4 MB
8192 MB
Processors:
2
1 CPU
4 CPUs
Enable EFI (special OSes only)
Help
Back
Next
Cancel


Create Virtual Machine
Virtual Hard disk
If you wish you can add a virtual hard disk to the new machine. You can either create a new hard disk file or select an existing one. Alternatively you can create a virtual machine without a virtual hard disk.
Create a Virtual Hard Disk Now
Disk Size:
20,00 GB
4.00 MB
2.00 TB
Pre-allocate Full Size
Use an Existing Virtual Hard Disk File
Ubuntu 20.04.vdi (Normal, 25.00 GB)
Do Not Add a Virtual Hard Disk
Help
Back
Next
Cancel



2.2)

```
vboxuser@KeenanBatista:~$ sudo adduser keenanbatista
Adding user `keenanbatista' ...
Adding new group `keenanbatista' (1001) ...
Adding new user `keenanbatista' (1001) with group `keenanbatista' ...
The home directory `/home/keenanbatista' already exists.  Not copying from `/etc/skel'.
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for keenanbatista
Enter the new value, or press ENTER for the default
        Full Name []: Keenan Batista
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
vboxuser@KeenanBatista:~$ sudo usermod -aG sudo keenanbatista
vboxuser@KeenanBatista:~$ groups keenanbatista
keenanbatista : keenanbatista sudo
vboxuser@KeenanBatista:~$
```

```
keenanbatista@KeenanBatista:~$ sudo usermod -l keenan_ubuntu vboxuser
keenanbatista@KeenanBatista:~$ sudo passwd keenan_ubuntu
New password:
Retype new password:
passwd: password updated successfully
keenanbatista@KeenanBatista:~$
```

2.3)

```
keenan_ubuntu@KeenanBatista:~$ sudo apt-get update
[sudo] password for keenan_ubuntu:
Hit:1 http://za.archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://za.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:3 http://za.archive.ubuntu.com/ubuntu focal-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu focal-security InRelease
Reading package lists... Done
keenan_ubuntu@KeenanBatista:~$ sudo apt-get upgrade
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following packages have been kept back:
  python3-update-manager ubuntu-advantage-tools update-manager update-manager-core
The following packages will be upgraded:
  accountsservice amd64-microcode apparmor apport apport-gtk apt apt-utils avahi-autoipd avahi-daemon avahi-utils base-files bind9-dnsutils
  bind9-host bind9-libs bluez bluez-cups bluez-obexd bolt bsdutils ca-certificates cpio cups cups-browsed cups-bsd cups-client cups-common
  cups-core-drivers cups-daemon cups-filters cups-filters-core-drivers cups-ipp-utils cups-ppdc cups-server-common distro-info
  distro-info-data dns-root-data dnsmasq-base fdisk firefox fonts-opensymbol fwupd fwupd-signed ghostscript ghostscript-x
  gir1.2-accountsservice-1.0 gir1.2-gdkpixbuf-2.0 gir1.2-gst-plugins-base-1.0 gir1.2-javascriptcoregtk-4.0 gir1.2-nm-1.0 gir1.2-rsvg-2.0
  gir1.2-vte-2.91 gir1.2-webkit2-4.0 gnome-control-center gnome-control-center-data gnome-control-center-faces gnome-shell
  gnome-shell-common gstreamer1.0-alsa gstreamer1.0-gl gstreamer1.0-gtk3 gstreamer1.0-plugins-base gstreamer1.0-plugins-base-apps
  gstreamer1.0-plugins-good gstreamer1.0-pulseaudio gstreamer1.0-x intel-microcode iptables iputils-ping iputils-tracepath klibc-utils
  krb5-locales less libaccountsservice0 libapparmor1 libapt-pkg6.0 libavahi-client3 libavahi-common-data libavahi-common3 libavahi-core7
  libavahi-glib1 libavahi-ui-gtk3-0 libblkid1 libbluetooth3 libc-bin libcap2 libcap2-bin libcue2 libcups2 libcupsfilters1 libcupsimage2
  libcurl3-gnutls libcurl4 libdw1 libelf1 libfdisk1 libflac8 libfontembed1 libfreerdp-client2-2 libfreerdp2-2 libfreetype6 libfwupd2
  libfwupdplugin5 libgdk-pixbuf2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7 libglib2.0-0 libglib2.0-bin libglib2.0-data
  libgnutls30 libgpgme11 libgpgmepp6 libgs9 libgs9-common libgssapi-krb5-2 libgstreamer-gl1.0-0 libgstreamer-plugins-base1.0-0
  libgstreamer-plugins-good1.0-0 libip4tc2 libip6tc2 libjavascriptcoregtk-4.0-18 libjuh-java libjurt-java libk5crypto3 libklibc libkpathsea6
  libkrb5-3 libkrb5support0 libldap-2.4-2 libldap-common libldb2 liblouis-data liblouis20 libmount1 libmysqlclient21 libncurses6
  libncursesw6 libndp0 libnetplan0 libnghttp2-14 libnm0 libnspr4 libnss-systemd libnss3 libpam-cap libpam-modules libpam-modules-bin
  libpam-runtime libpam-systemd libpam0g libperl5.30 libpoppler-cpp0v5 libpoppler-glib8 libpoppler97 libprocps8 libpython3.8
```

```
  libpython3.8-minimal libpython3.8-stdlib libraw19 libreoffice-base-core libreoffice-calc libreoffice-common libreoffice-core
  libreoffice-draw libreoffice-gnome libreoffice-gtk3 libreoffice-impress libreoffice-math libreoffice-ogltrans libreoffice-pdfimport
  libreoffice-style-breeze libreoffice-style-colibre libreoffice-style-elementary libreoffice-style-tango libreoffice-writer libridl-java
  librsvg2-2 librsvg2-common libsmartcols1 libsmbclient libsndfile1 libsnmp-base libsnmp35 libsqlite3-0 libssh-4 libssl1.1 libsynctex2
  libsystemd0 libtiff5 libtinfo6 libtss2-esys0 libuno-cppu3 libuno-cppuhelpergcc3-3 libuno-purpenvhelpergcc3-3 libuno-sal3
  libuno-salhelpergcc3-3 libunoloader-java libunwind8 libuuid1 libuv1 libvpx6 libvte-2.91-0 libvte-2.91-common libwbclient0
  libwebkit2gtk-4.0-37 libwebp6 libwebpdemux2 libwebpmux3 libwinpr2-2 libx11-6 libx11-data libx11-xcb1 libxml2 libxpm4 libxtables12 libyajl2
  linux-firmware login ltrace mount ncurses-base ncurses-bin netplan.io network-manager network-manager-config-connectivity-ubuntu
  openssh-client openssl openvpn passwd perl perl-base perl-modules-5.30 poppler-utils printer-driver-foo2zjs printer-driver-foo2zjs-common
  printer-driver-splix procps python3-apport python3-cryptography python3-debian python3-distro-info python3-idna python3-ldb python3-louis
  python3-paramiko python3-pil python3-problem-report python3-renderpm python3-reportlab python3-reportlab-accel python3-requests
  python3-software-properties python3-uno python3-urllib3 python3.8 python3.8-minimal remmina remmina-common remmina-plugin-rdp
  remmina-plugin-secret remmina-plugin-vnc rfkill rsync samba-libs snapd software-properties-common software-properties-gtk sudo systemd
  systemd-sysv systemd-timesyncd tar tcpdump thunderbird thunderbird-gnome-support thunderbird-locale-en tracker-extract tracker-miner-fs
  tzdata ubuntu-drivers-common ufw uno-libs-private unzip update-notifier update-notifier-common ure util-linux uuid-runtime vim-common
  vim-tiny xserver-common xserver-xephyr xserver-xorg-core xserver-xorg-legacy xwayland xxd
296 upgraded, 0 newly installed, 0 to remove and 4 not upgraded.
Need to get 71.1 MB/506 MB of archives.
After this operation, 20.3 MB disk space will be freed.
Do you want to continue? [Y/n] Y
```

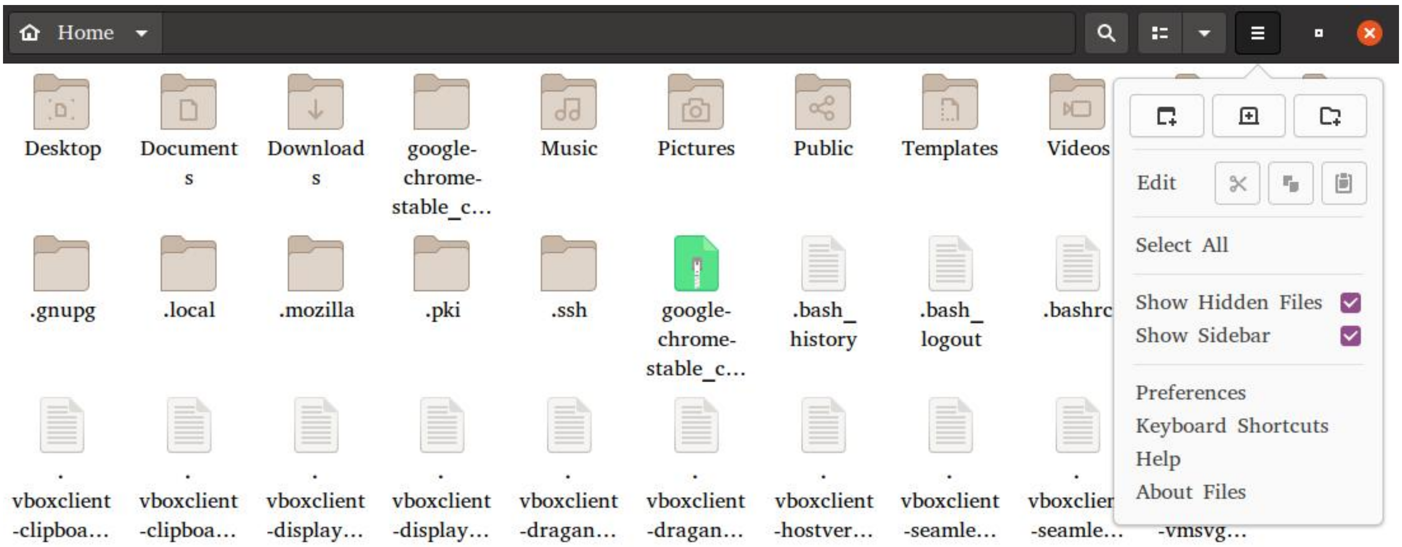
```
keenan_ubuntu@KeenanBatista:~$ wget https://dl.google.com/linux/direct/google-chrome-stable_current_amd64.deb
--2024-06-16 00:40:57--  https://dl.google.com/linux/direct/google-chrome-stable_current_amd64.deb
Resolving dl.google.com (dl.google.com)... 192.178.54.14, 2c0f:fb50:4002:812::200e
Connecting to dl.google.com (dl.google.com)|192.178.54.14|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 108800244 (104M) [application/x-debian-package]
Saving to: ‘google-chrome-stable_current_amd64.deb’

google-chrome-stable_current_amd64. 100%[=================================================================>] 103.76M  2.03MB/s    in 60s

2024-06-16 00:42:04 (1.72 MB/s) - ‘google-chrome-stable_current_amd64.deb’ saved [108800244/108800244]

keenan_ubuntu@KeenanBatista:~$ sudo dpkg -i google-chrome-stable_current_amd64.deb
Selecting previously unselected package google-chrome-stable.
(Reading database ... 184644 files and directories currently installed.)
Preparing to unpack google-chrome-stable_current_amd64.deb ...
Unpacking google-chrome-stable (126.0.6478.61-1) ...
Setting up google-chrome-stable (126.0.6478.61-1) ...
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/x-www-browser (x-www-browser) in auto mode
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/gnome-www-browser (gnome-www-browser) in auto mode
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/google-chrome (google-chrome) in auto mode
Processing triggers for gnome-menus (3.36.0-1ubuntu1) ...
Processing triggers for desktop-file-utils (0.24-1ubuntu3) ...
Processing triggers for mime-support (3.64ubuntu1) ...
Processing triggers for man-db (2.9.1-1) ...
keenan_ubuntu@KeenanBatista:~$
```

```
keenan_ubuntu@keenanbatista:~$ ls -a
.
..
.bash_history
.bash_logout
.bashrc
.cache
.config
Desktop
Documents
Downloads
.gnupg
google-chrome-stable_current_amd64
google-chrome-stable_current_amd64.deb
.local
.mozilla
Music
Pictures
.pki
.profile
Public
.ssh
.sudo_as_admin_successful
Templates
.vboxclient-clipboard-tty2-control.pid
```



```
keenan_ubuntu@KeenanBatista:~$ hostnamectl set-hostname keenanbatista
keenan_ubuntu@KeenanBatista:~$
```

```
keenan_ubuntu@KeenanBatista:~$ cd Documents
keenan_ubuntu@KeenanBatista:~/Documents$ mkdir WorkDocuments
keenan_ubuntu@KeenanBatista:~/Documents$ cd WorkDocuments
keenan_ubuntu@KeenanBatista:~/Documents/WorkDocuments$ touch textfile.txt
keenan_ubuntu@KeenanBatista:~/Documents/WorkDocuments$
```

```
keenan_ubuntu@keenanbatista:~/Documents/WorkDocuments$ cat textfile.txt
```

2.4)

```
keenan_ubuntu@keenanbatista:~$ sudo ufw status
Status: inactive
keenan_ubuntu@keenanbatista:~$ sudo ufw reset
Resetting all rules to installed defaults. Proceed with operation (y|n)? y
Backing up 'user.rules' to '/etc/ufw/user.rules.20240616_013740'
Backing up 'before.rules' to '/etc/ufw/before.rules.20240616_013740'
Backing up 'after.rules' to '/etc/ufw/after.rules.20240616_013740'
Backing up 'user6.rules' to '/etc/ufw/user6.rules.20240616_013740'
Backing up 'before6.rules' to '/etc/ufw/before6.rules.20240616_013740'
Backing up 'after6.rules' to '/etc/ufw/after6.rules.20240616_013740'

keenan_ubuntu@keenanbatista:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
keenan_ubuntu@keenanbatista:~$ sudo ufw deny ssh
Rules updated
Rules updated (v6)
keenan_ubuntu@keenanbatista:~$ sudo ufw deny http
Rules updated
Rules updated (v6)
keenan_ubuntu@keenanbatista:~$ sudo ufw deny https
Rules updated
Rules updated (v6)
keenan_ubuntu@keenanbatista:~$ sudo ufw enable
Firewall is active and enabled on system startup
```

```
keenan_ubuntu@keenanbatista:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     DENY        Anywhere
80/tcp                     DENY        Anywhere
443/tcp                    DENY        Anywhere
22/tcp (v6)                DENY        Anywhere (v6)
80/tcp (v6)                DENY        Anywhere (v6)
443/tcp (v6)               DENY        Anywhere (v6)
```

Question 3

3.1)

```
keenan_ubuntu@keenanbatista:~$ sudo apt install ubuntu-gnome-desktop
[sudo] password for keenan_ubuntu:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  adwaita-icon-theme-full fonts-cantarell gnome-session
The following NEW packages will be installed:
  adwaita-icon-theme-full fonts-cantarell gnome-session ubuntu-gnome-desktop
0 upgraded, 4 newly installed, 0 to remove and 4 not upgraded.
Need to get 7,203 kB of archives.
After this operation, 22.9 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://za.archive.ubuntu.com/ubuntu focal-updates/universe amd64 adwaita-icon-theme-full all 3.36.1-2ubuntu0.20.04.2 [6,936 kB]
Get:2 http://za.archive.ubuntu.com/ubuntu focal/universe amd64 fonts-cantarell all 0.111-2 [250 kB]
Get:3 http://za.archive.ubuntu.com/ubuntu focal/universe amd64 gnome-session all 3.36.0-2ubuntu1 [13.4 kB]
Get:4 http://za.archive.ubuntu.com/ubuntu focal/universe amd64 ubuntu-gnome-desktop amd64 0.87 [3,100 B]
Fetched 7,203 kB in 38s (191 kB/s)
Selecting previously unselected package adwaita-icon-theme-full.
(Reading database ... 184757 files and directories currently installed.)
Preparing to unpack .../adwaita-icon-theme-full_3.36.1-2ubuntu0.20.04.2_all.deb ...
Unpacking adwaita-icon-theme-full (3.36.1-2ubuntu0.20.04.2) ...
Selecting previously unselected package fonts-cantarell.
Preparing to unpack .../fonts-cantarell_0.111-2_all.deb ...
Unpacking fonts-cantarell (0.111-2) ...
Selecting previously unselected package gnome-session.
Preparing to unpack .../gnome-session_3.36.0-2ubuntu1_all.deb ...
Unpacking gnome-session (3.36.0-2ubuntu1) ...
Selecting previously unselected package ubuntu-gnome-desktop.
Preparing to unpack .../ubuntu-gnome-desktop_0.87_amd64.deb ...
Unpacking ubuntu-gnome-desktop (0.87) ...
Setting up fonts-cantarell (0.111-2) ...
```

```
Setting up adwaita-icon-theme-full (3.36.1-2ubuntu0.20.04.2) ...
Setting up gnome-session (3.36.0-2ubuntu1) ...
Setting up ubuntu-gnome-desktop (0.87) ...
Processing triggers for fontconfig (2.13.1-2ubuntu3) ...
```

3.2)

```
keenan_ubuntu@keenanbatista:~$ sudo apt install gnome-shell-extensions
```

```
keenan_ubuntu@keenanbatista:~$ sudo apt install gnome-tweak-tool
[sudo] password for keenan_ubuntu:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  gnome-tweak-tool
0 upgraded, 1 newly installed, 0 to remove and 8 not upgraded.
Need to get 1,848 B of archives.
After this operation, 12.3 kB of additional disk space will be used.
Get:1 http://za.archive.ubuntu.com/ubuntu focal/universe amd64 gnome-tweak-tool all 3.34.0-2ubuntu1 [1,848 B]
Fetched 1,848 B in 1s (1,737 B/s)
Selecting previously unselected package gnome-tweak-tool.
(Reading database ... 189233 files and directories currently installed.)
Preparing to unpack .../gnome-tweak-tool_3.34.0-2ubuntu1_all.deb ...
Unpacking gnome-tweak-tool (3.34.0-2ubuntu1) ...
Setting up gnome-tweak-tool (3.34.0-2ubuntu1) ...
```

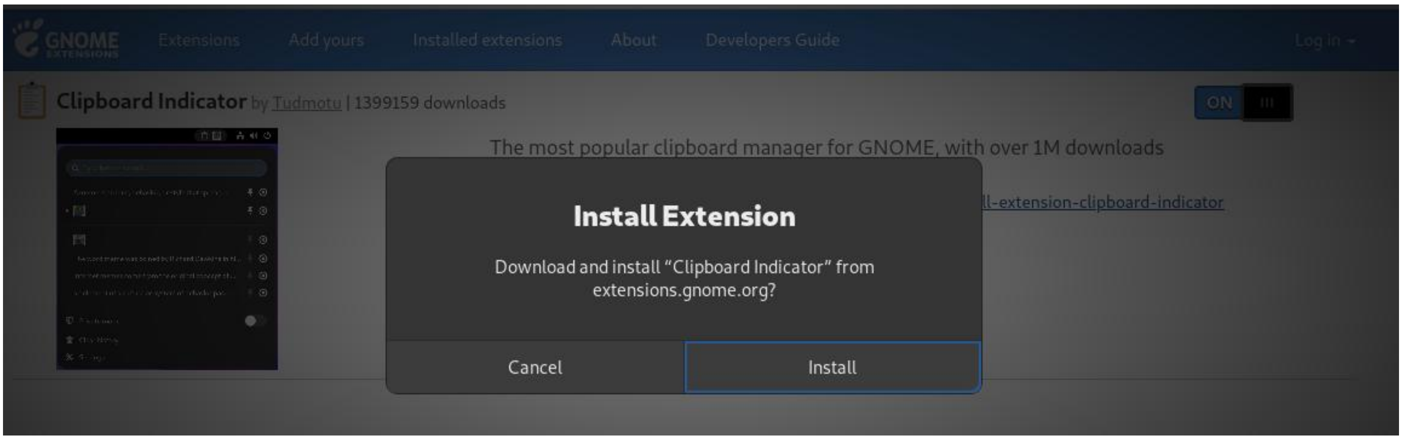
```
keenan_ubuntu@keenanbatista:~$ sudo apt install chrome-gnome-shell
[sudo] password for keenan_ubuntu:
Reading package lists... Done
Building dependency tree
Reading state information... Done
chrome-gnome-shell is already the newest version (10.1-5).
chrome-gnome-shell set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 8 not upgraded.
```

Allow extensions.gnome.org to install an add-on?

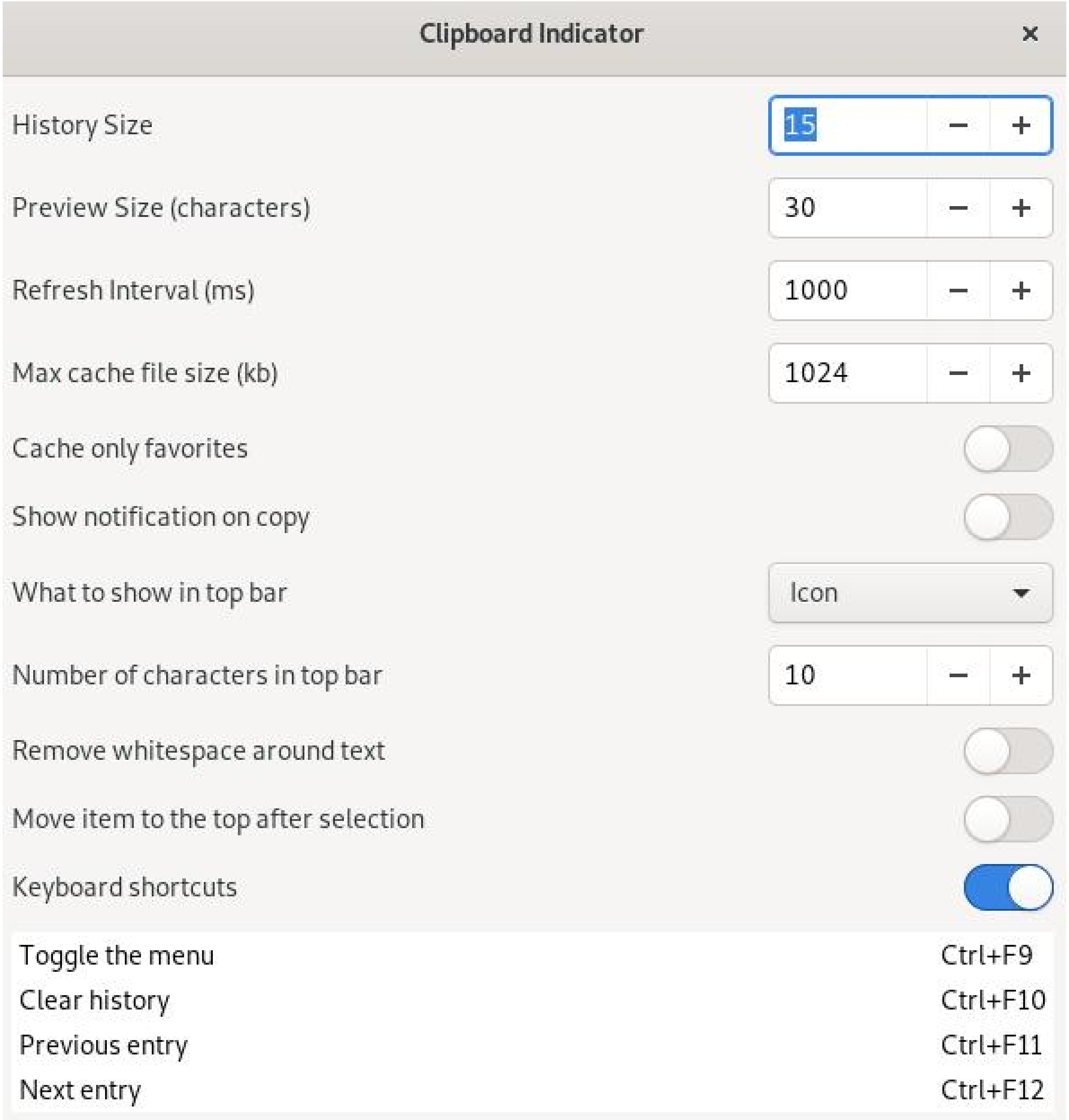
You are attempting to install an add-on from extensions.gnome.org. Make sure you trust this site before continuing.

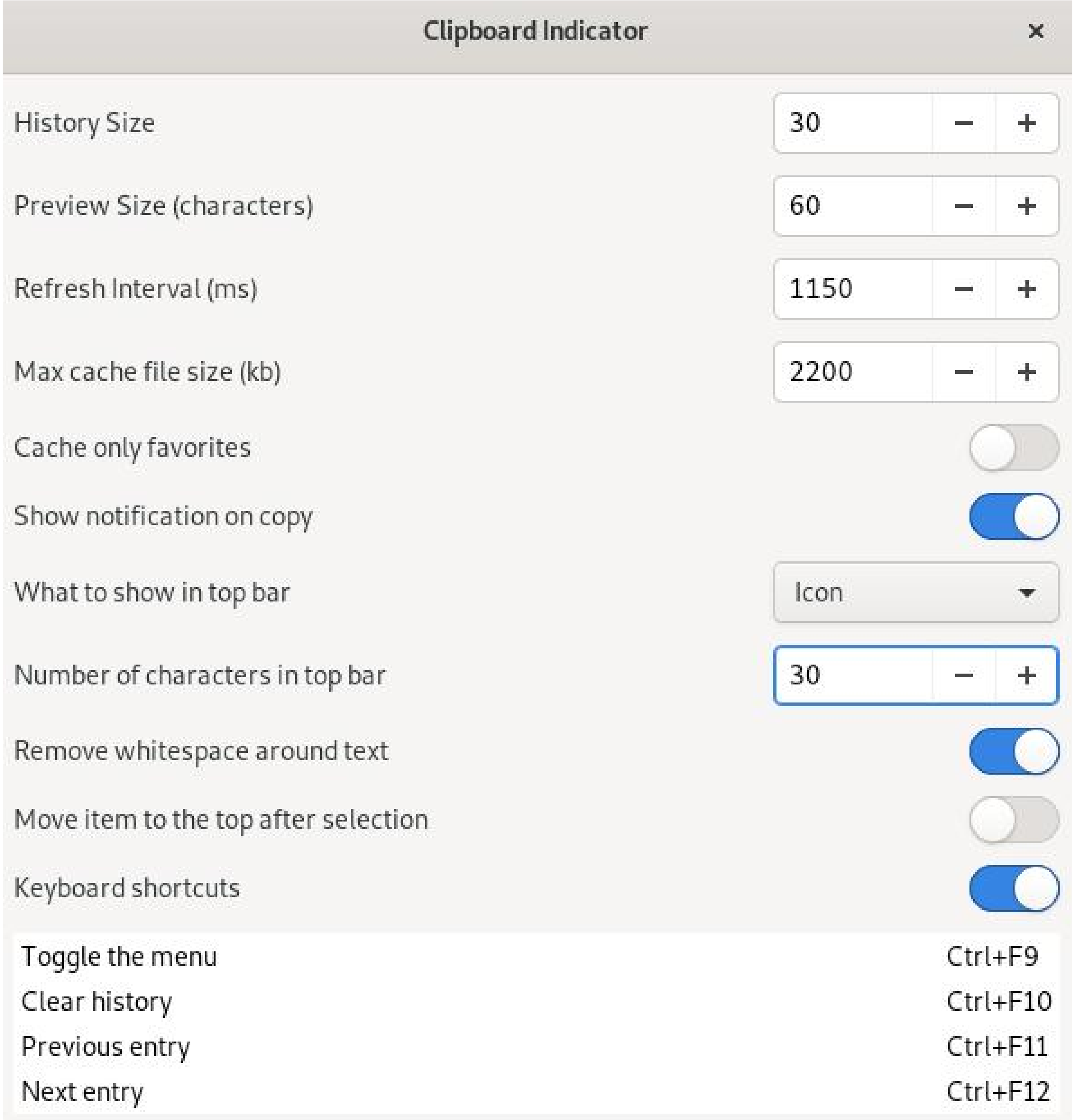
Learn more about installing add-ons safely

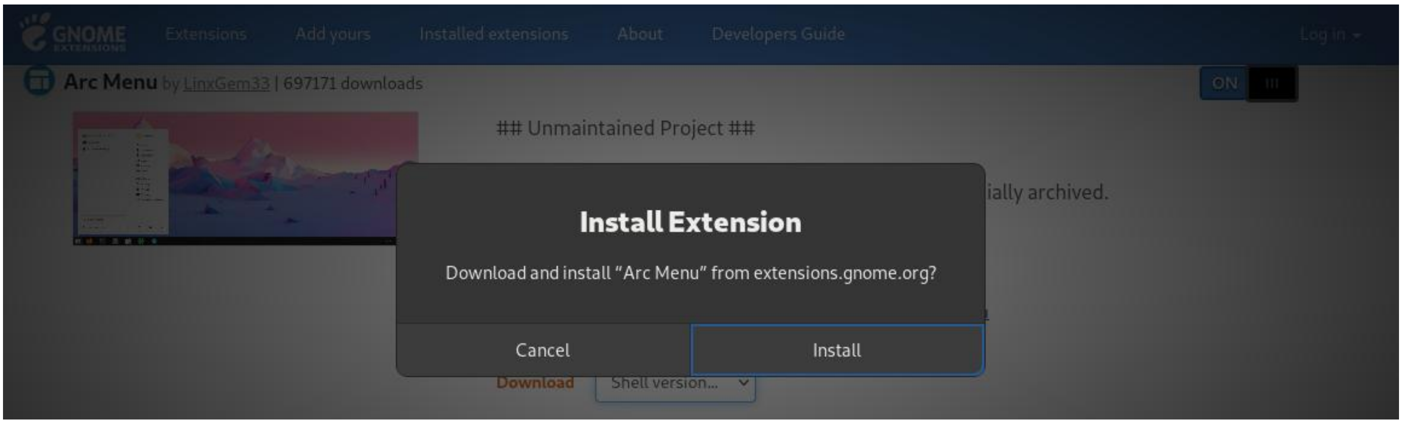
Don't Allow | Continue to Installation



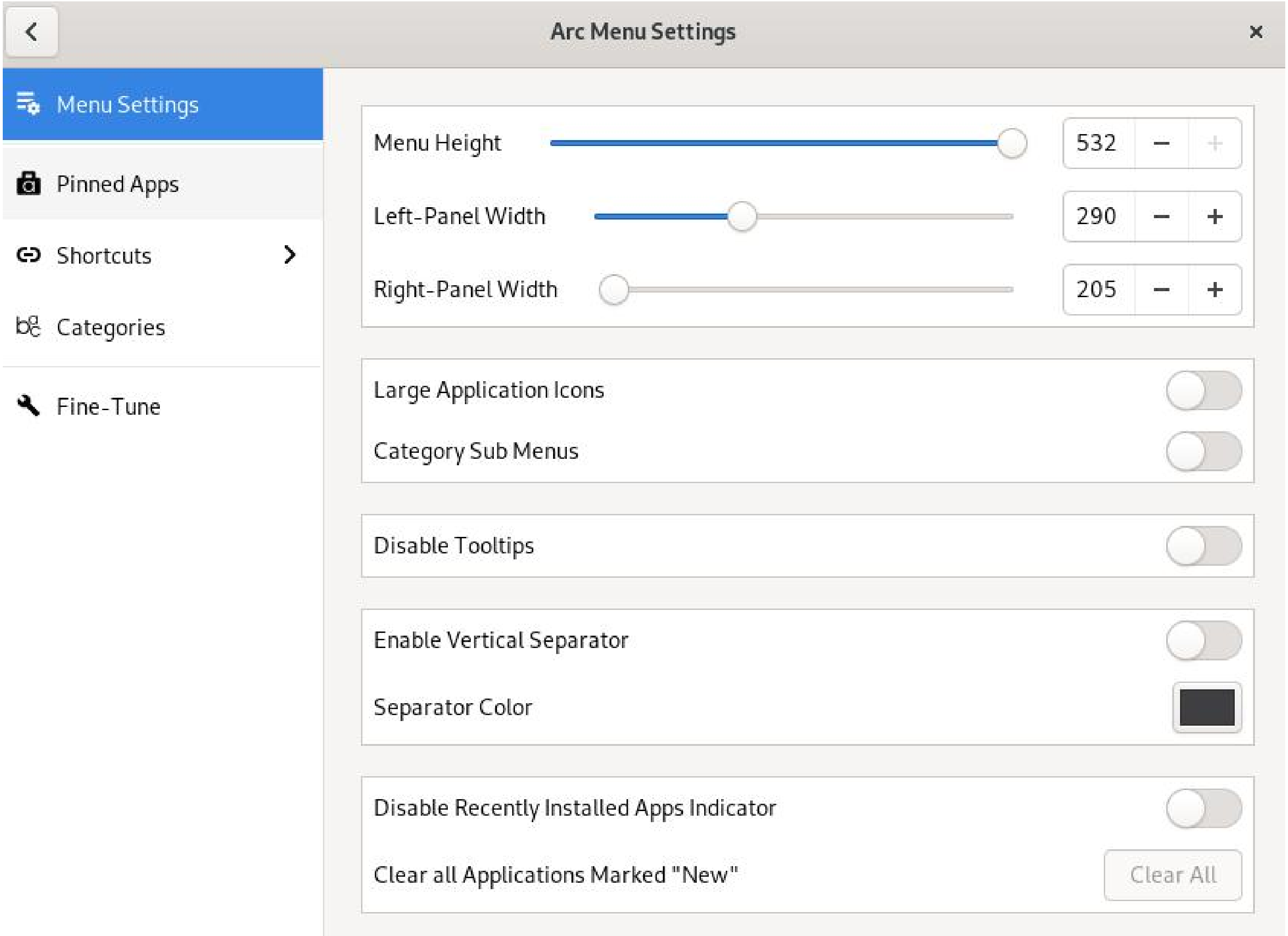
Clipboard indicator
Clipboard manager extension for gnome-shell - adds a clipboard indicator to the top panel, and caches clipboard history.

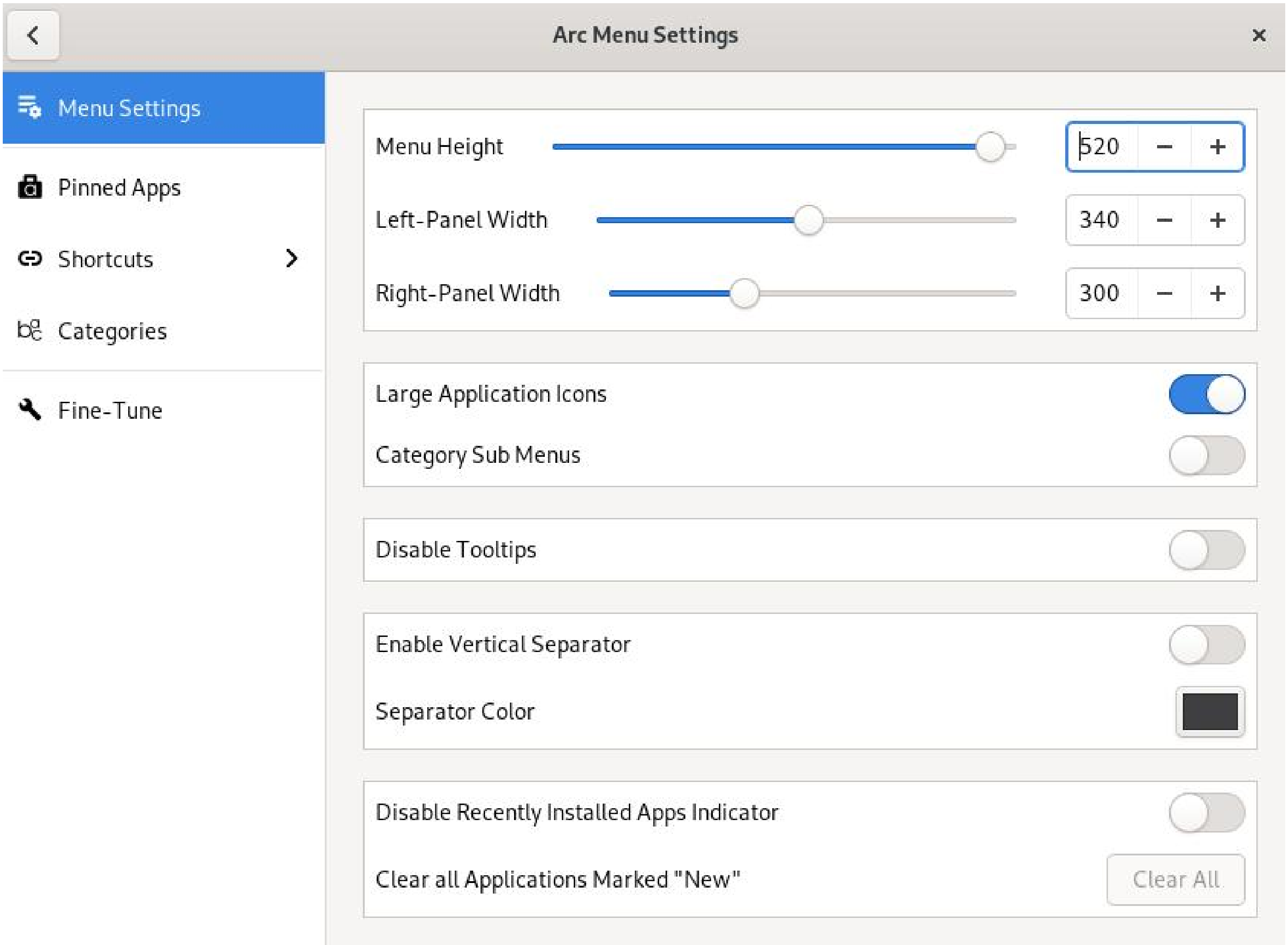

Clipboard Indicator
History Size
15
Preview Size (characters)
30
Refresh Interval (ms)
1000
Max cache file size (kb)
1024
Cache only favorites
Show notification on copy
What to show in top bar
Icon
Number of characters in top bar
10
Remove whitespace around text
Move item to the top after selection
Keyboard shortcuts
Toggle the menu
Ctrl+F9
Clear history
Ctrl+F10
Previous entry
Ctrl+F11
Next entry
Ctrl+F12


Clipboard Indicator
History Size
30
Preview Size (characters)
60
Refresh Interval (ms)
1150
Max cache file size (kb)
2200
Cache only favorites
Show notification on copy
What to show in top bar
Icon
Number of characters in top bar
30
Remove whitespace around text
Move item to the top after selection
Keyboard shortcuts
Toggle the menu
Ctrl+F9
Clear history
Ctrl+F10
Previous entry
Ctrl+F11
Next entry
Ctrl+F12


Arc Menu by LinxGem33 | 697171 downloads
Unmaintained Project
ially archived.
Install Extension
Download and install “Arc Menu” from extensions.gnome.org?
Cancel
Install
Download


Arc menu
unmaintained project


Arc Menu Settings
Menu Settings
Pinned Apps
Shortcuts
Categories
Fine-Tune
Menu Height
532
Left-Panel Width
290
Right-Panel Width
205
Large Application Icons
Category Sub Menus
Disable Tooltips
Enable Vertical Separator
Separator Color
Disable Recently Installed Apps Indicator
Clear all Applications Marked "New"
Clear All

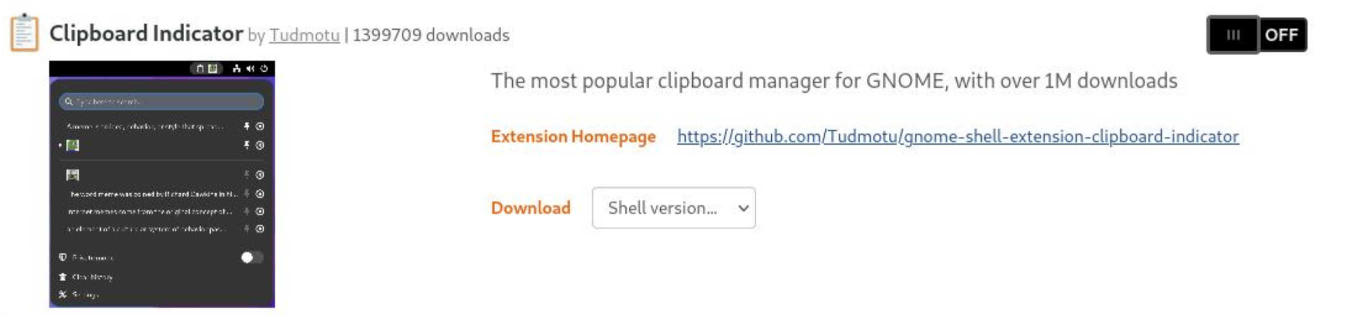

Arc Menu Settings
Menu Settings
Pinned Apps
Shortcuts
Categories
Fine-Tune
Menu Height
520
Left-Panel Width
340
Right-Panel Width
300
Large Application Icons
Category Sub Menus
Disable Tooltips
Enable Vertical Separator
Separator Color
Disable Recently Installed Apps Indicator
Clear all Applications Marked "New"
Clear All

3.3) Clipboard indicator is an extension that stores a history of clipboard data, which allows users to re-copy it at a later time. This will increase the productivity of the employees in TechCorp as any important text or links that are used regularly can be saved to the clipboard indicator and copied by the employee instead of having to retype the entire text or link.

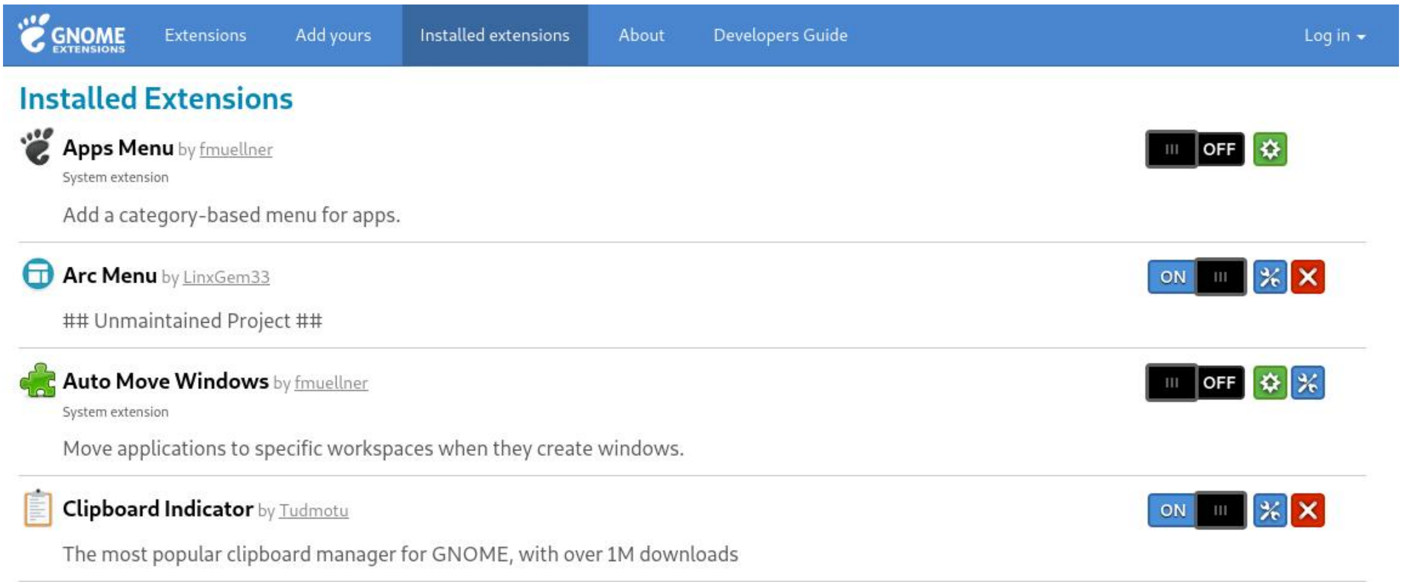
Arc menu is an extension that provides a more traditional application menu akin to the start menu of Windows and is highly customisable and configurable. Like the clipboard indicator extension, arc menu increases the productivity of TechCorp's employees the menu provides fast and easy access to important programs and applications like terminal, settings, Gnome Tweak Tool, and the main folders of file manager such as Home, Documents and Downloads. It can further increase employee productivity as the employees are able to configure it to what suits them best.

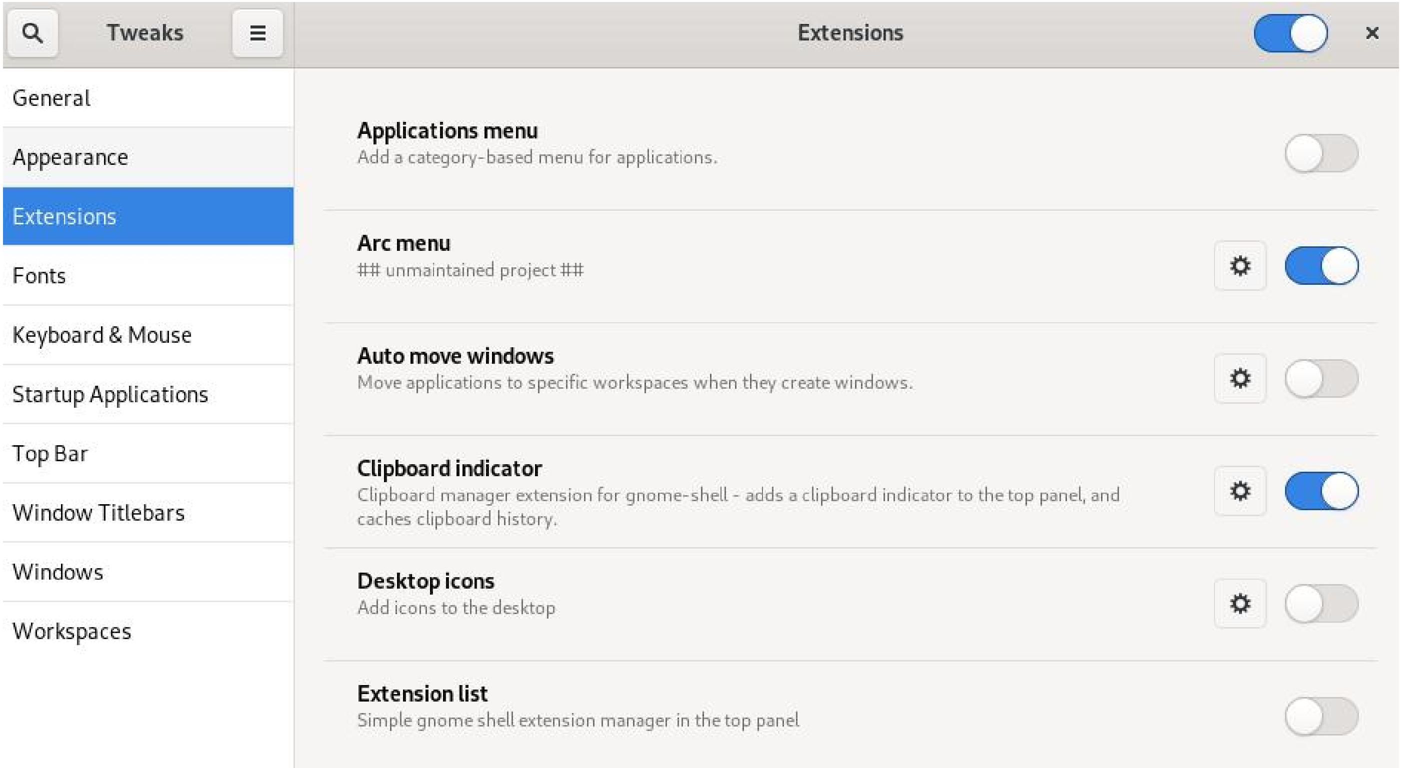
In order to install both extensions, the user must enter the GNOME Shell Extensions website on the browser of their choice and must search for the extension using the search function that the website provides.

After finding the extension, the user must then enter it and click the switch from off to on to install the extension.

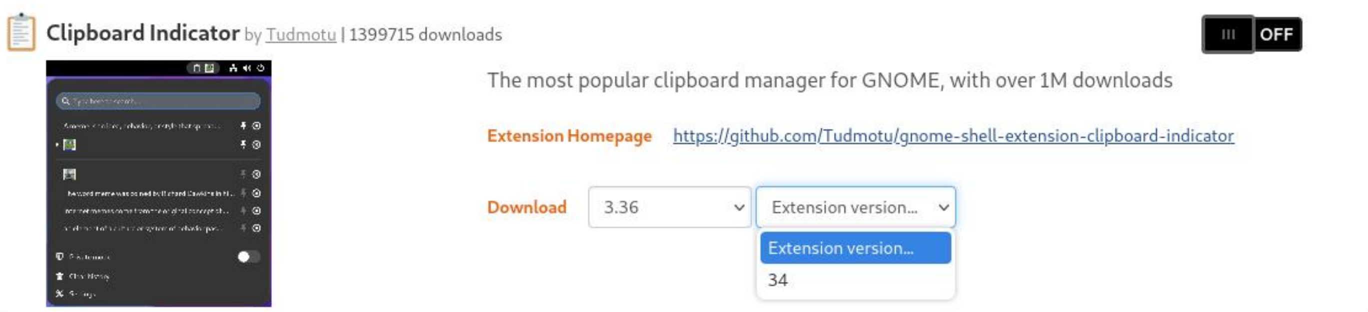


You can find the downloaded extension on either the GNOME Shell Extension website or Gnome Tweak Tool.

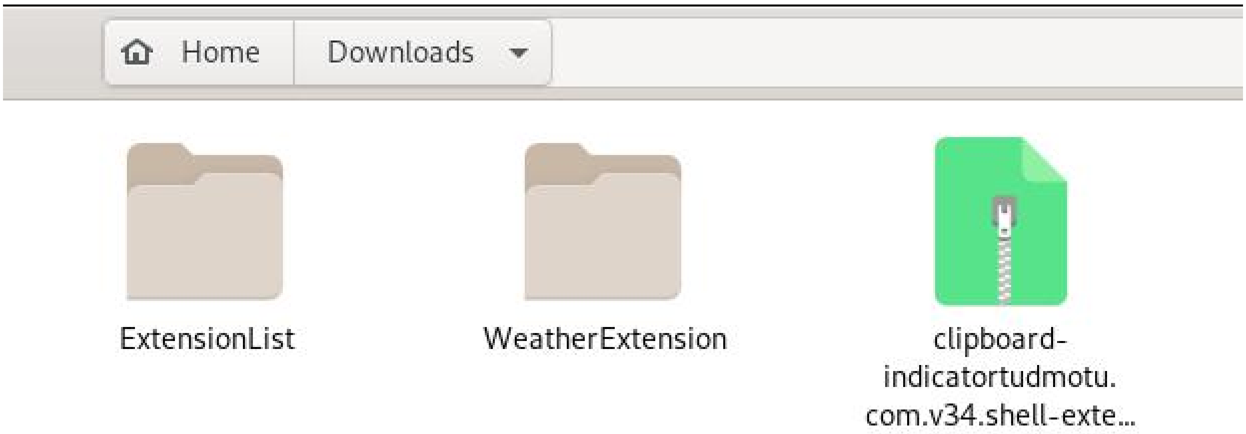


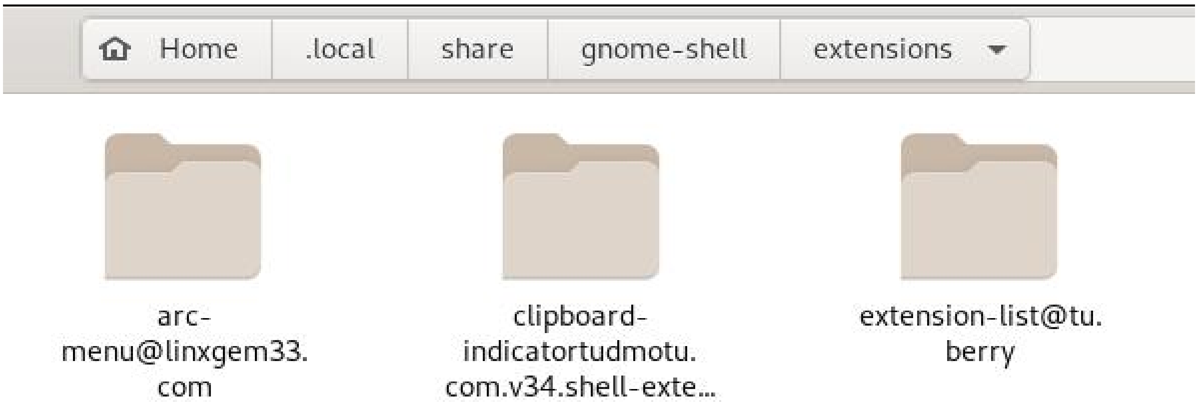


If the user wishes to manually install the extension, then they must click the shell version drop box and choose the shell and extension version and it will automatically install after that.

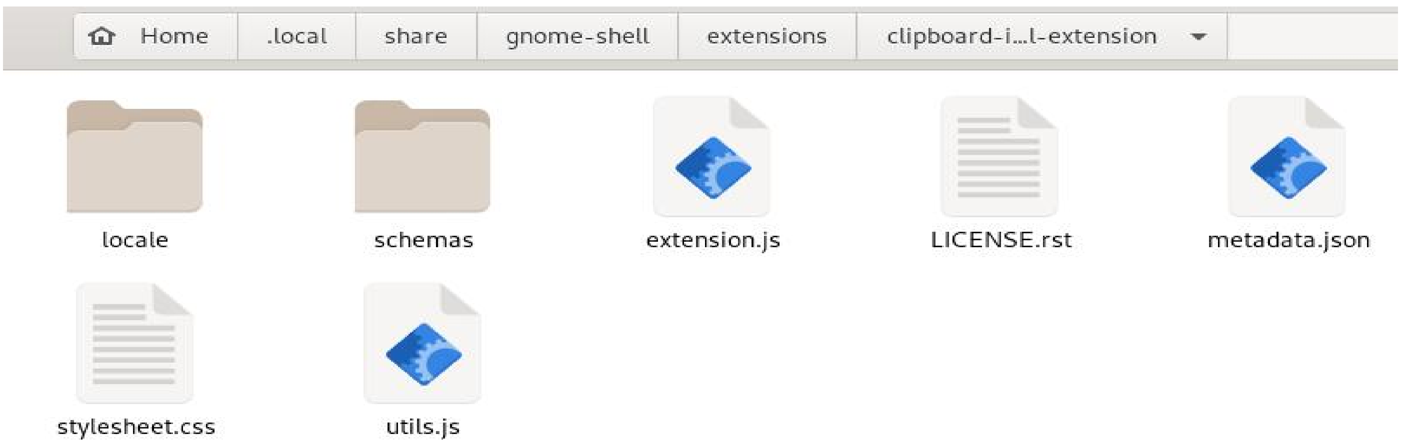


The zip folder of the extension is found in the Downloads folder and the user must extract it to the folder at Home/.local/share/gnome-shell/extensions (The .local folder is a hidden file and can be seen by using ctrl + H).





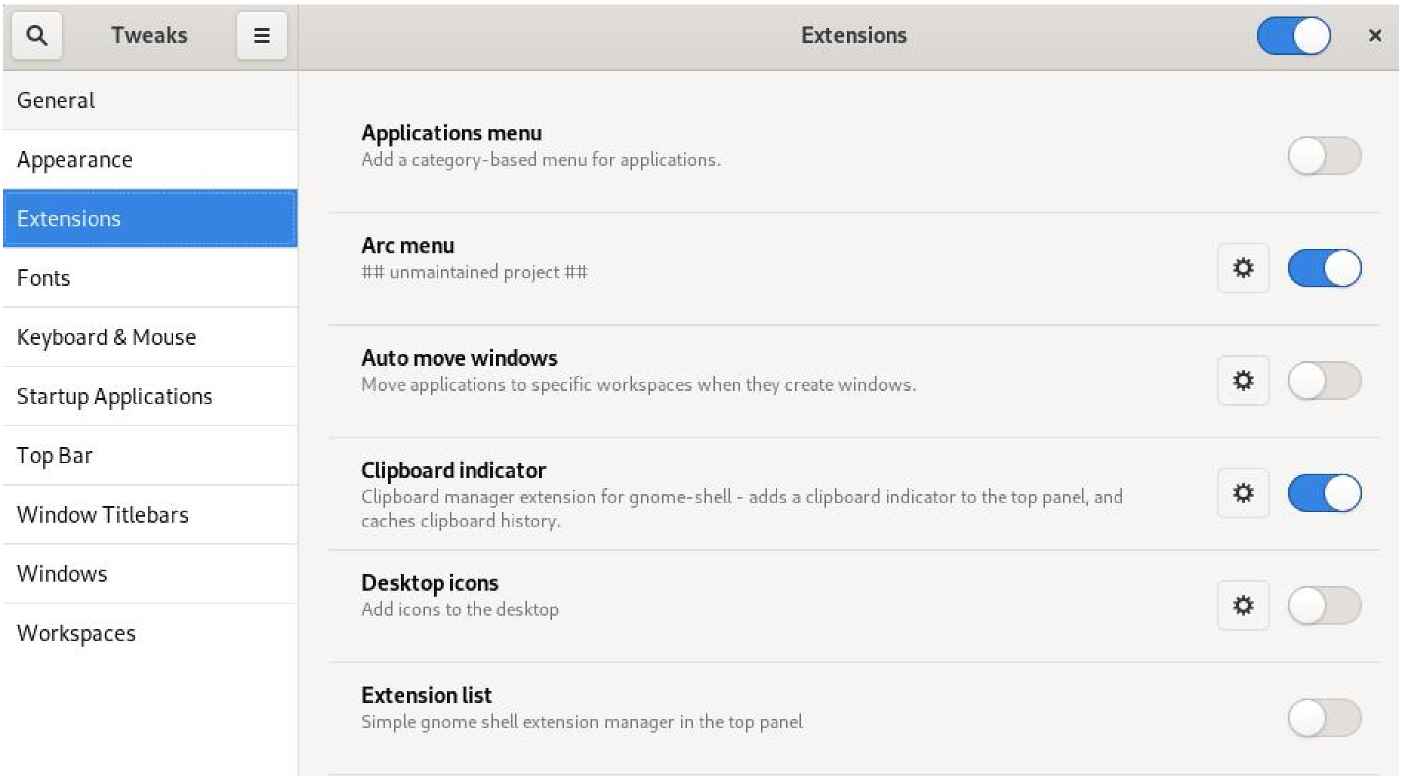
In order for Gnome Tweak Tool to recognise the extension you must rename the extension folder to its uuid that is found in the metadata.json file in the same extension folder.



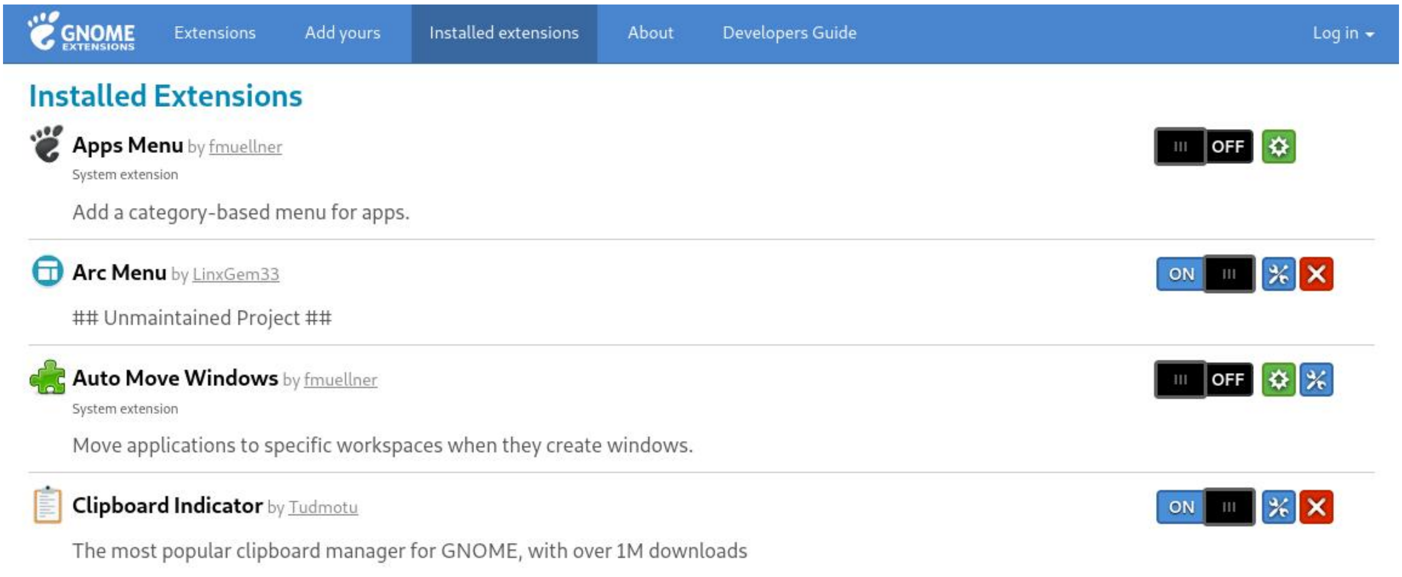
After this the Gnome shell must restart in order for it to recognise the new extension. This can be done by restarting the machine or using alt + f2 and typing “r” into the command line (Prakash, 2023).

You can configure these extensions with either the Gnome Tweak Tool or on the GNOME Shell Extensions website.

Configuring extensions with Gnome Tweak Tool is done by opening the application, entering the extensions sub menu, and clicking on the cog icon of the extension the user would like to configure.

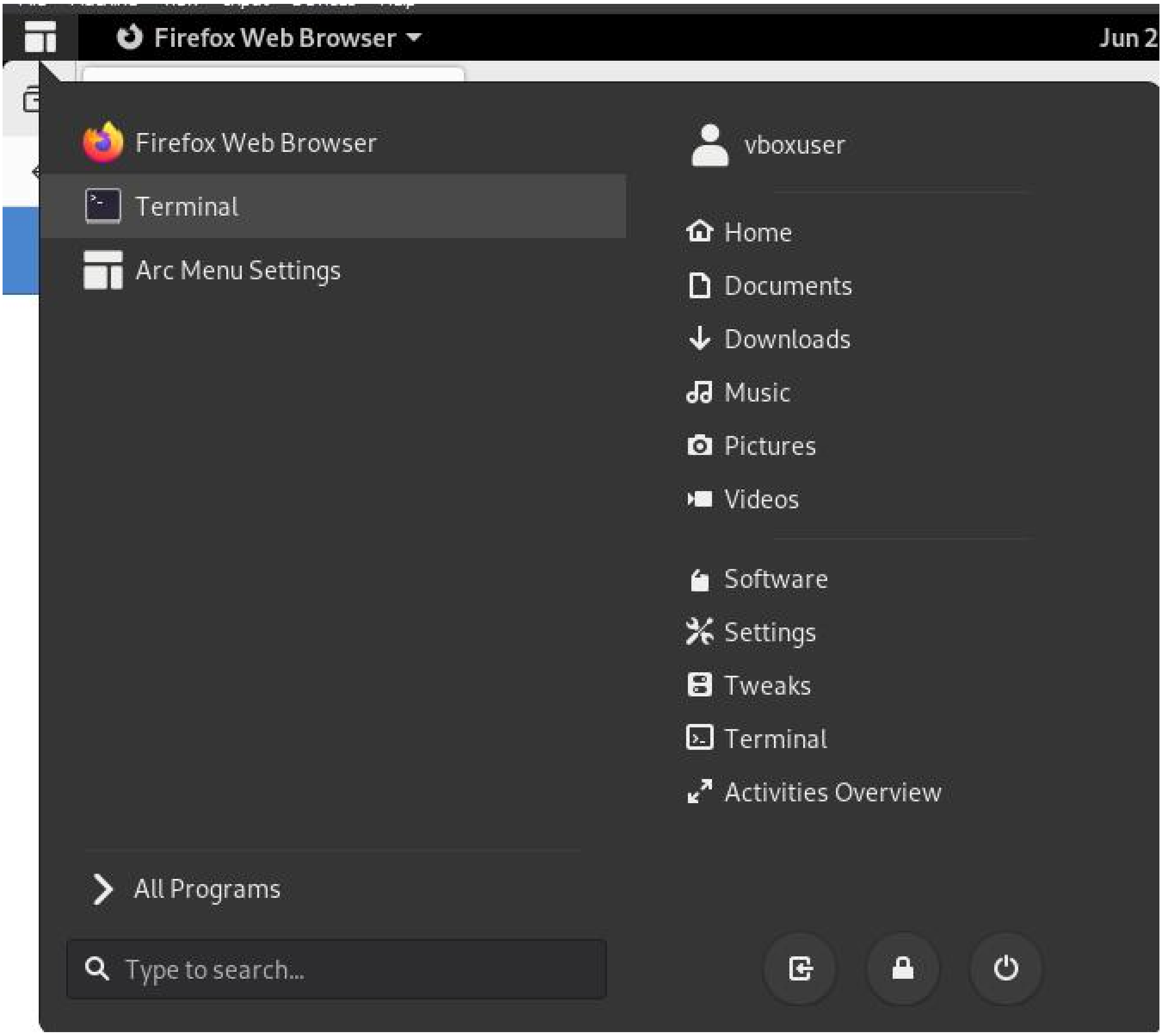


Configuring extensions on the GNOME Shell Extensions website is entering the website on any browser, entering the “Installed extensions” sub menu, and clicking on the icon with the screwdriver and wrench of the extension the user would like to configure.

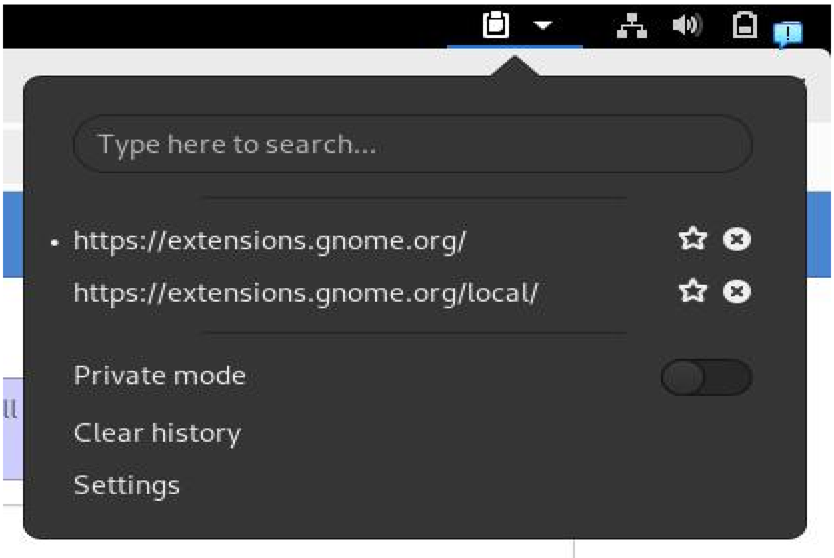


The following screenshots displays the extensions working after installation.

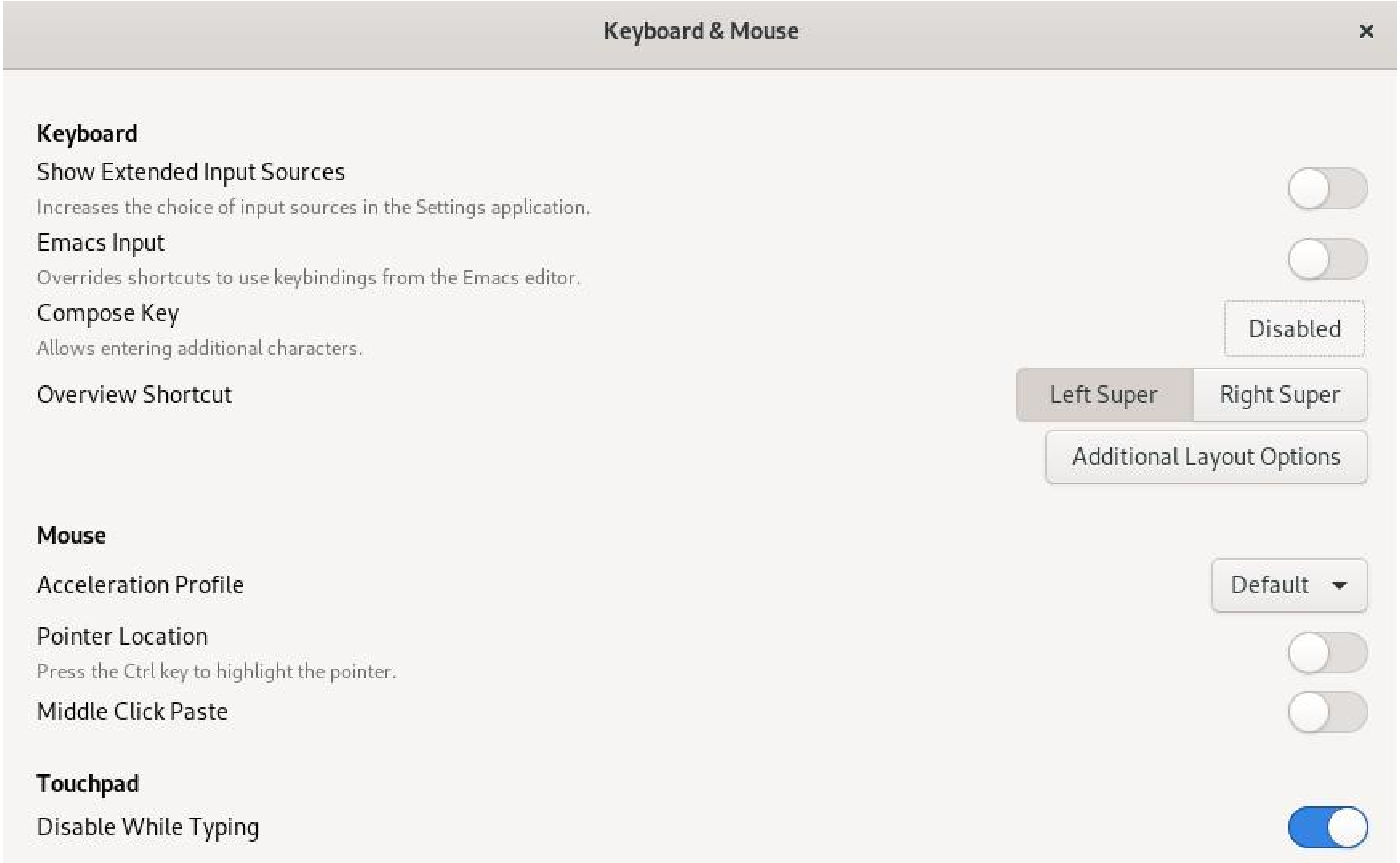
Arc menu extension:



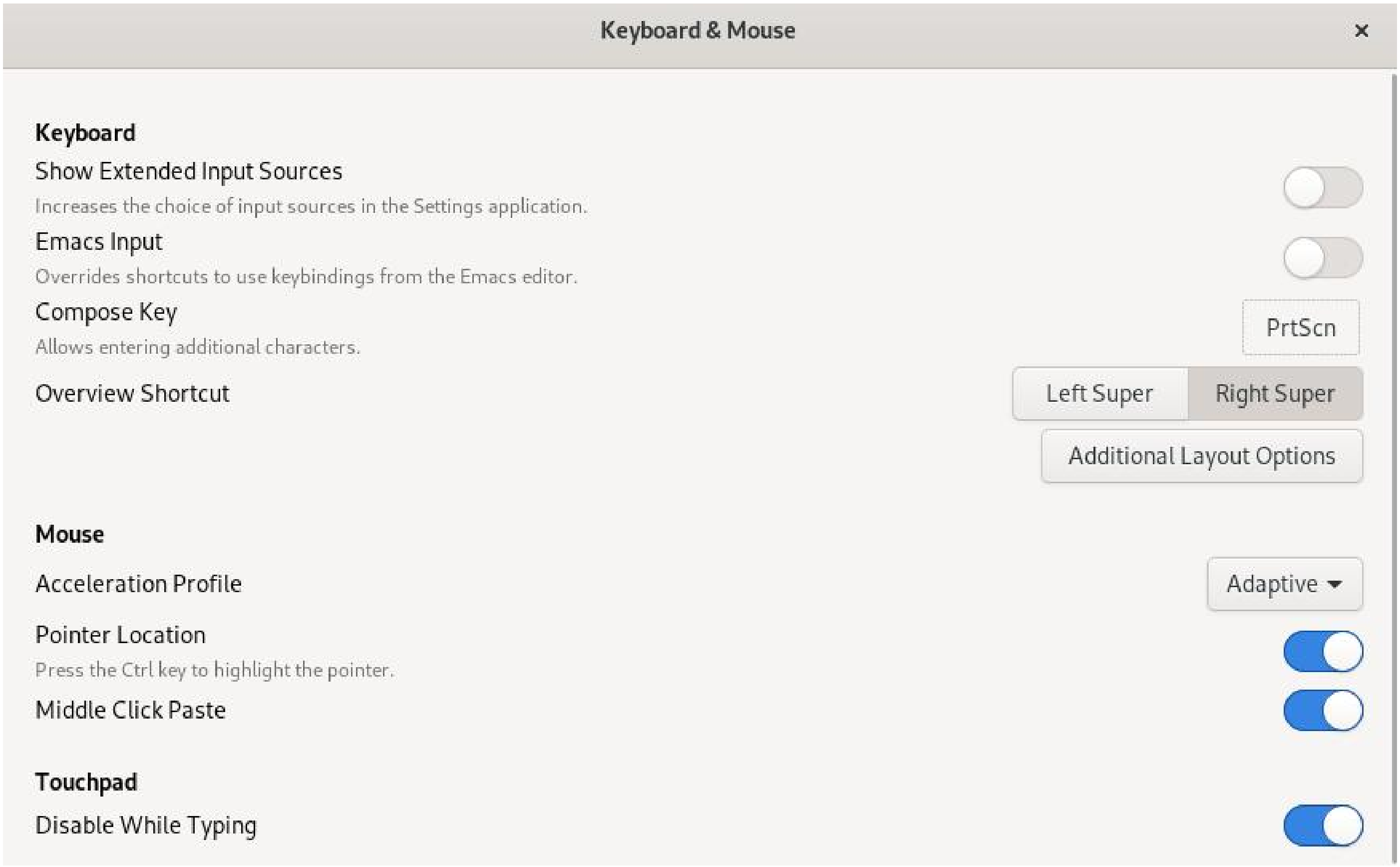
Clipboard indicator extension:



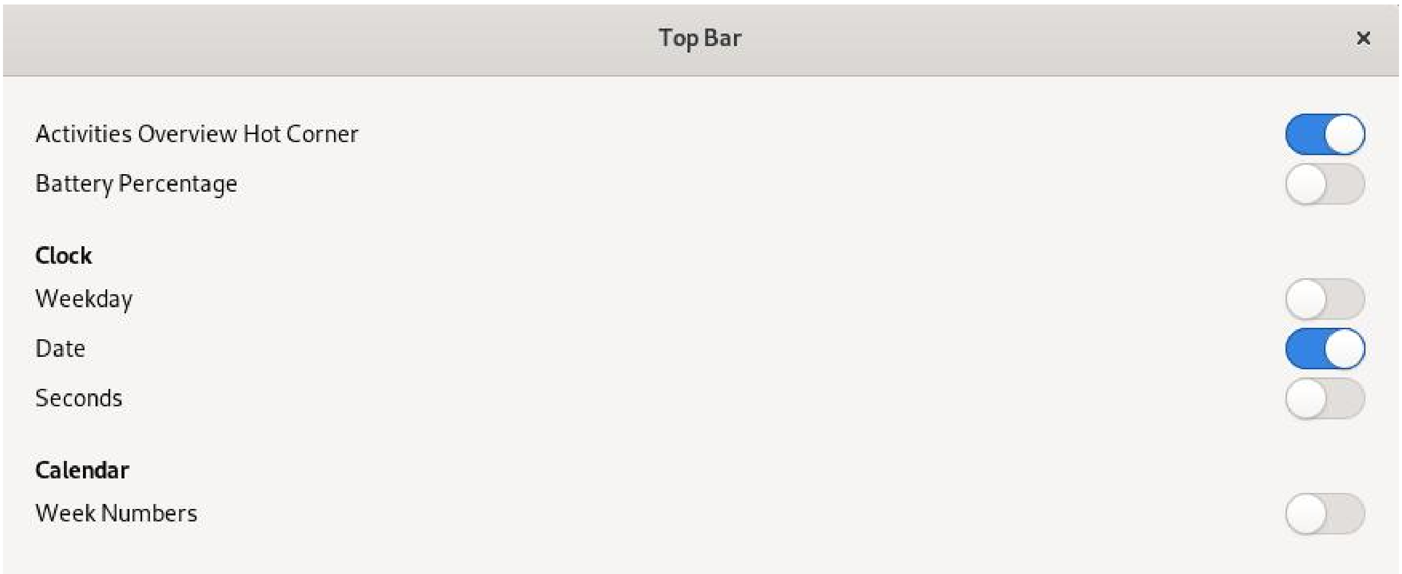
3.4.1)



Keyboard & Mouse
Keyboard
Show Extended Input Sources
Increases the choice of input sources in the Settings application.
Emacs Input
Overrides shortcuts to use keybindings from the Emacs editor.
Compose Key
Allows entering additional characters.
Disabled
Overview Shortcut
Left Super
Right Super
Additional Layout Options
Mouse
Acceleration Profile
Default
Pointer Location
Press the Ctrl key to highlight the pointer.
Middle Click Paste
Touchpad
Disable While Typing



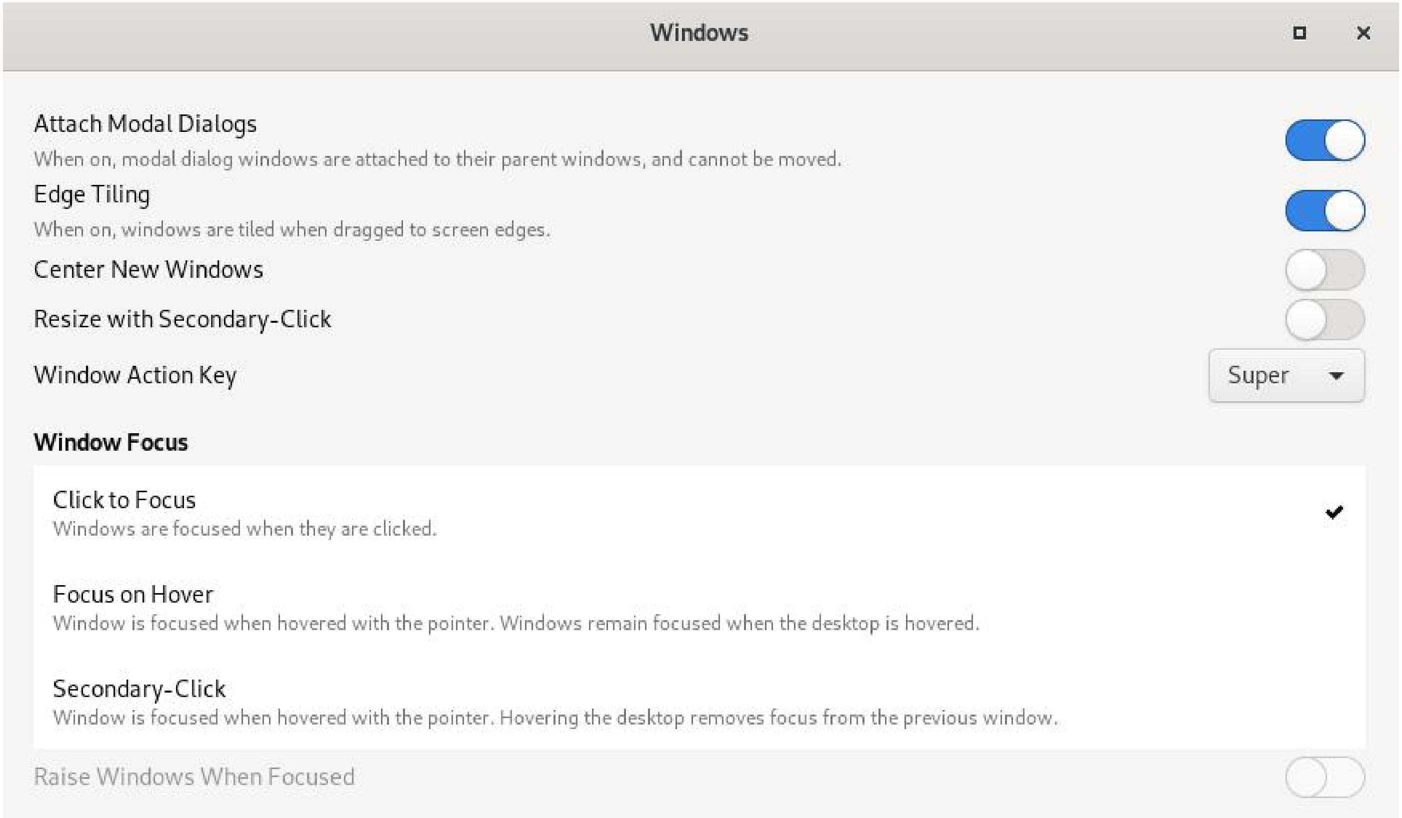
Keyboard & Mouse
Keyboard
Show Extended Input Sources
Increases the choice of input sources in the Settings application.
Emacs Input
Overrides shortcuts to use keybindings from the Emacs editor.
Compose Key
Allows entering additional characters.
PrtScn
Overview Shortcut
Left Super
Right Super
Additional Layout Options
Mouse
Acceleration Profile
Adaptive
Pointer Location
Press the Ctrl key to highlight the pointer.
Middle Click Paste
Touchpad
Disable While Typing

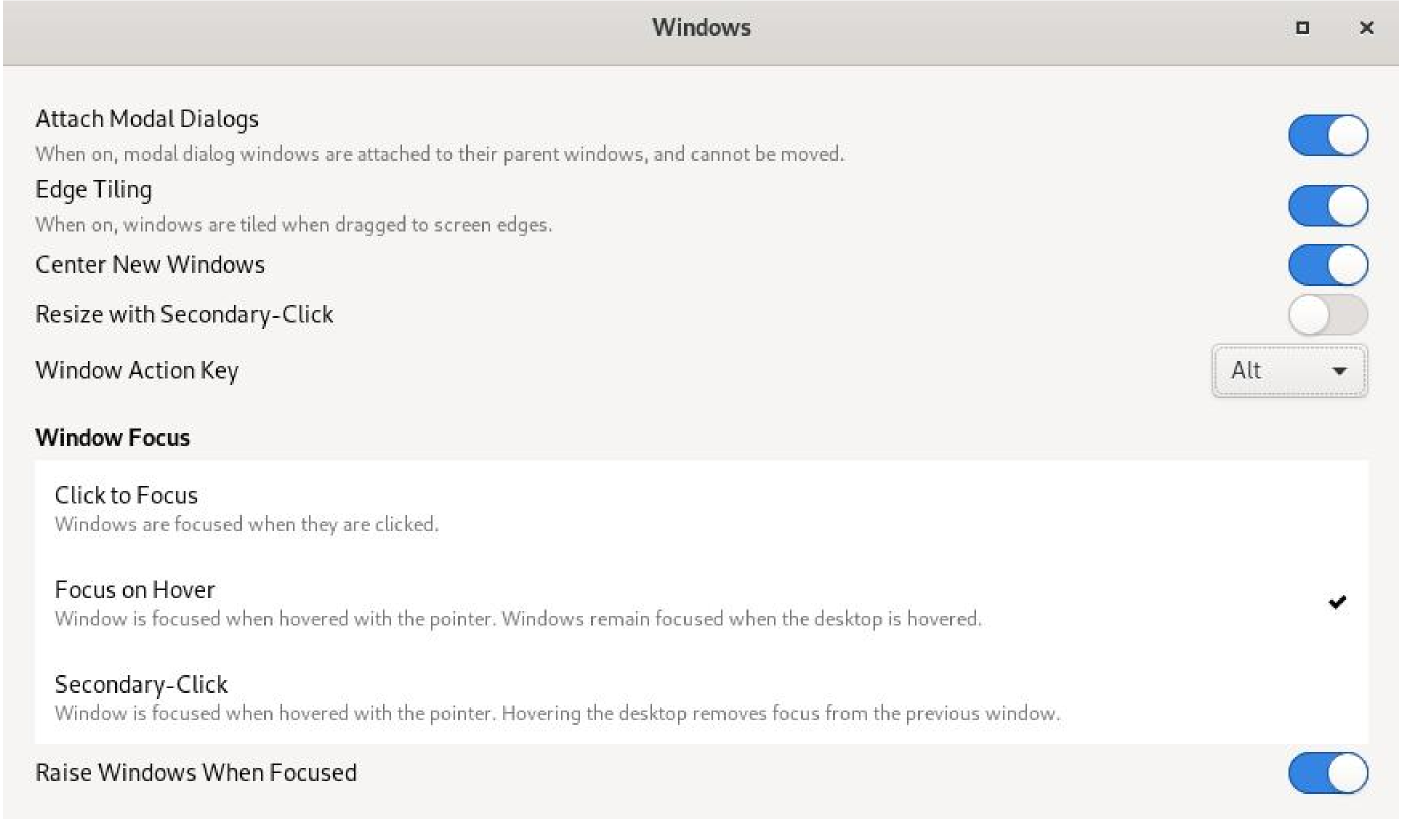

Top Bar
Activities Overview Hot Corner
Battery Percentage
Clock
Weekday
Date
Seconds
Calendar
Week Numbers

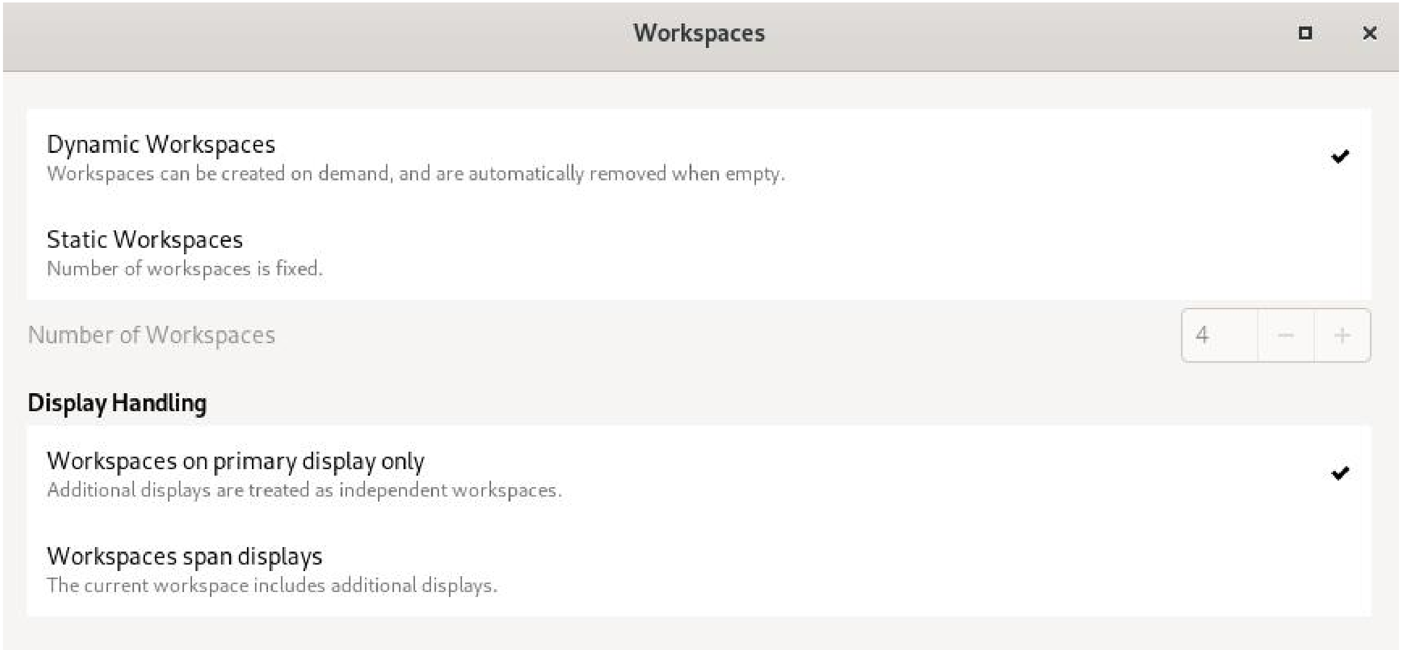

Top Bar
Activities Overview Hot Corner
Battery Percentage
Clock
Weekday
Date
Seconds
Calendar
Week Numbers

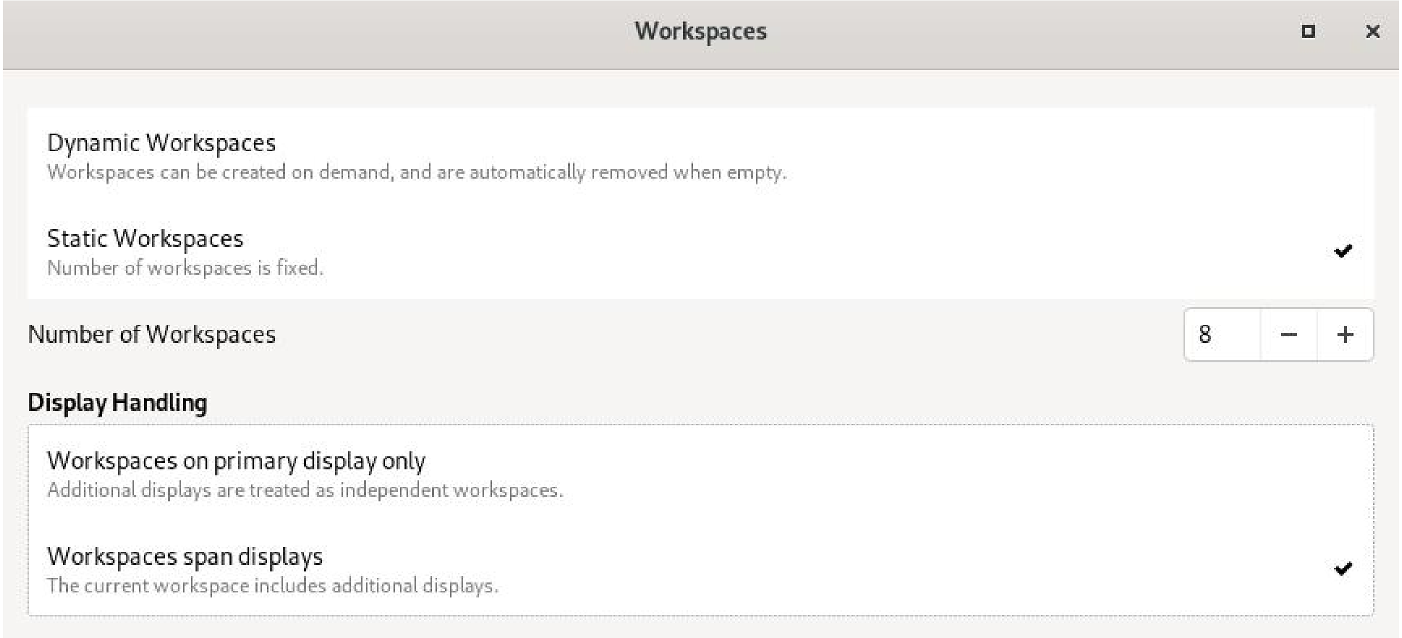

Window Titlebars
Titlebar Actions
Double-Click
Toggle Maximize
Middle-Click
None
Secondary-Click
Menu
Titlebar Buttons
Maximize
Minimize
Placement
Left
Right


Window Titlebars
Titlebar Actions
Double-Click
Toggle Maximize
Middle-Click
Menu
Secondary-Click
Minimize
Titlebar Buttons
Maximize
Minimize
Placement
Left
Right


Windows
Attach Modal Dialogs
When on, modal dialog windows are attached to their parent windows, and cannot be moved.
Edge Tiling
When on, windows are tiled when dragged to screen edges.
Center New Windows
Resize with Secondary-Click
Window Action Key
Super
Window Focus
Click to Focus
Windows are focused when they are clicked.
Focus on Hover
Window is focused when hovered with the pointer. Windows remain focused when the desktop is hovered.
Secondary-Click
Window is focused when hovered with the pointer. Hovering the desktop removes focus from the previous window.
Raise Windows When Focused


Windows
Attach Modal Dialogs
When on, modal dialog windows are attached to their parent windows, and cannot be moved.
Edge Tiling
When on, windows are tiled when dragged to screen edges.
Center New Windows
Resize with Secondary-Click
Window Action Key
Alt
Window Focus
Click to Focus
Windows are focused when they are clicked.
Focus on Hover
Window is focused when hovered with the pointer. Windows remain focused when the desktop is hovered.
Secondary-Click
Window is focused when hovered with the pointer. Hovering the desktop removes focus from the previous window.
Raise Windows When Focused


Workspaces
Dynamic Workspaces
Workspaces can be created on demand, and are automatically removed when empty.
Static Workspaces
Number of workspaces is fixed.
Number of Workspaces
4
Display Handling
Workspaces on primary display only
Additional displays are treated as independent workspaces.
Workspaces span displays
The current workspace includes additional displays.


Workspaces
Dynamic Workspaces
Workspaces can be created on demand, and are automatically removed when empty.
Static Workspaces
Number of workspaces is fixed.
Number of Workspaces
8
Display Handling
Workspaces on primary display only
Additional displays are treated as independent workspaces.
Workspaces span displays
The current workspace includes additional displays.

3.4.2)

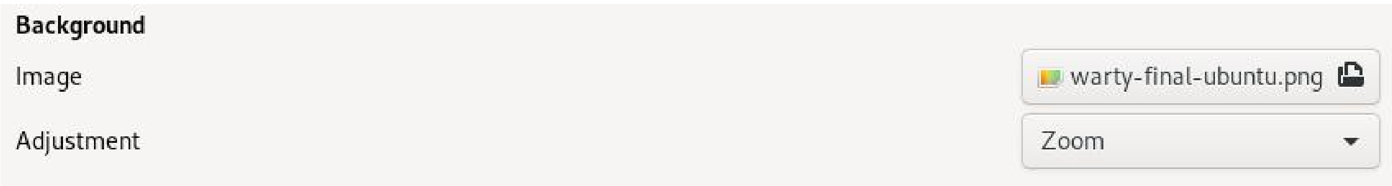
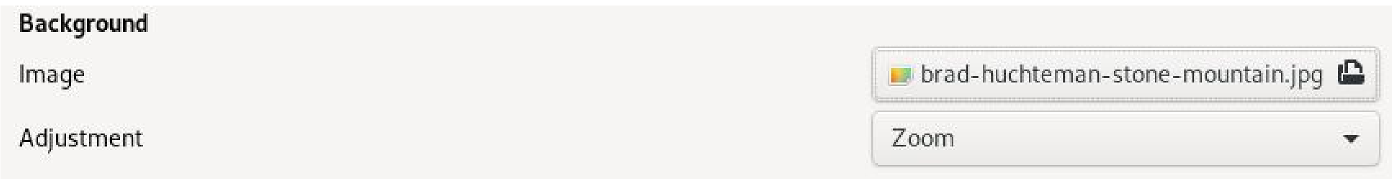
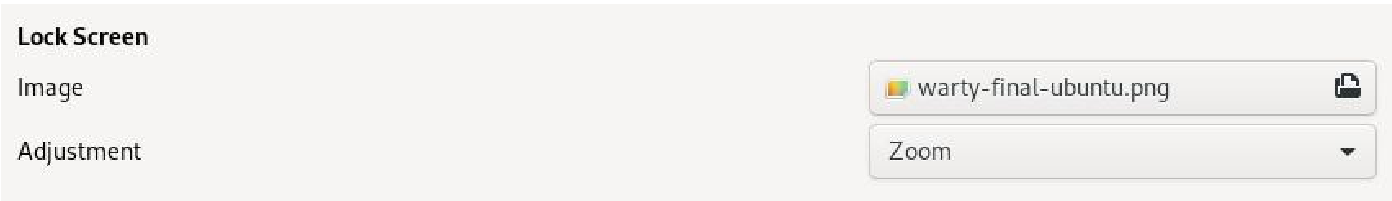

Background
Image
warty-final-ubuntu.png
Adjustment
Zoom


Image
Open
usr
share
backgrounds
Name
Size
Type
Modified
contest
16 Mar 2023
brad-huchteman-stone-mountain.jpg 1.9 MB Image 2 Apr 2020
Focal-Fossa_WP_4096x2304_GREY.png 2.0 MB Image 2 Apr 2020
hardy_wallpaper_uhd.png 1.4 MB Image 2 Apr 2020
joshua-coleman-something-yellow.jpg 3.1 MB Image 2 Apr 2020
matt-mcnulty-nyc-2nd-ave.jpg 1.4 MB Image 2 Apr 2020
ryan-stone-skykomish-river.jpg 2.2 MB Image 2 Apr 2020
ubuntu-default-greyscale-wallpaper.png 2.0 MB Image 2 Apr 2020
warty-final-ubuntu.png 2.2 MB Image 16 Mar 2023


Background
Image
brad-huchteman-stone-mountain.jpg
Adjustment
Zoom

3.4.3)


Lock Screen
Image
warty-final-ubuntu.png
Adjustment
Zoom

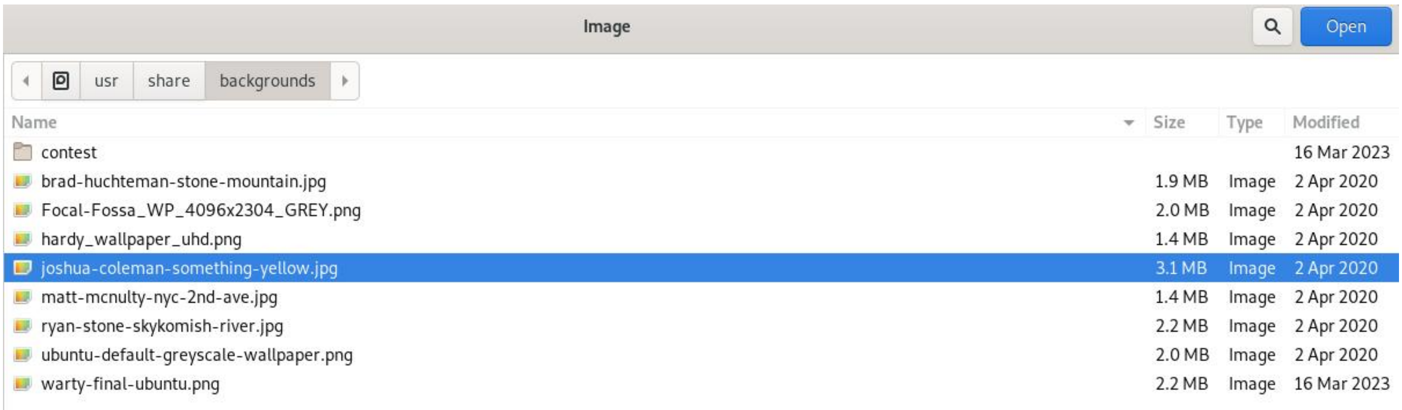
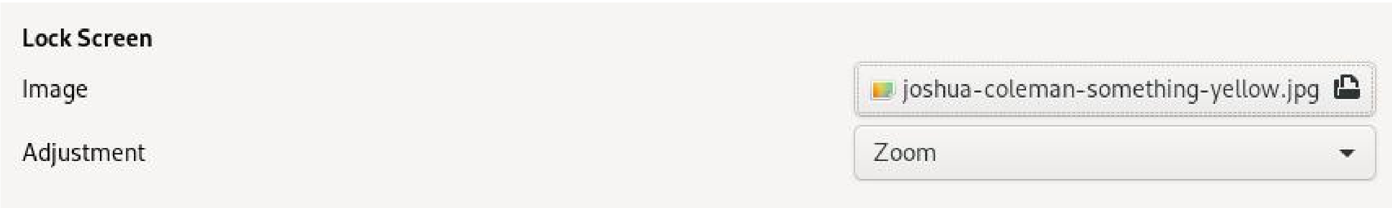
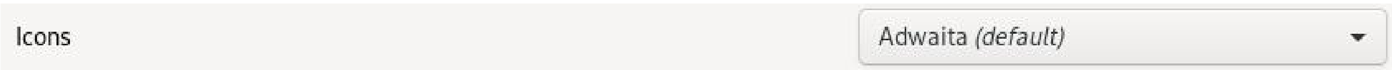
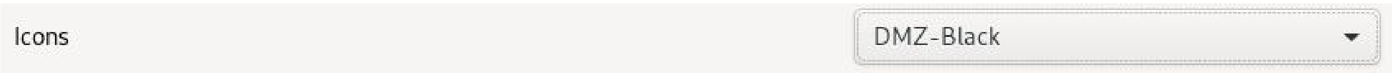


Image
Open
usr
share
backgrounds
Name
Size
Type
Modified
contest 16 Mar 2023
brad-huchteman-stone-mountain.jpg 1.9 MB Image 2 Apr 2020
Focal-Fossa_WP_4096x2304_GREY.png 2.0 MB Image 2 Apr 2020
hardy_wallpaper_uhd.png 1.4 MB Image 2 Apr 2020
joshua-coleman-something-yellow.jpg 3.1 MB Image 2 Apr 2020
matt-mcnulty-nyc-2nd-ave.jpg 1.4 MB Image 2 Apr 2020
ryan-stone-skykomish-river.jpg 2.2 MB Image 2 Apr 2020
ubuntu-default-greyscale-wallpaper.png 2.0 MB Image 2 Apr 2020
warty-final-ubuntu.png 2.2 MB Image 16 Mar 2023

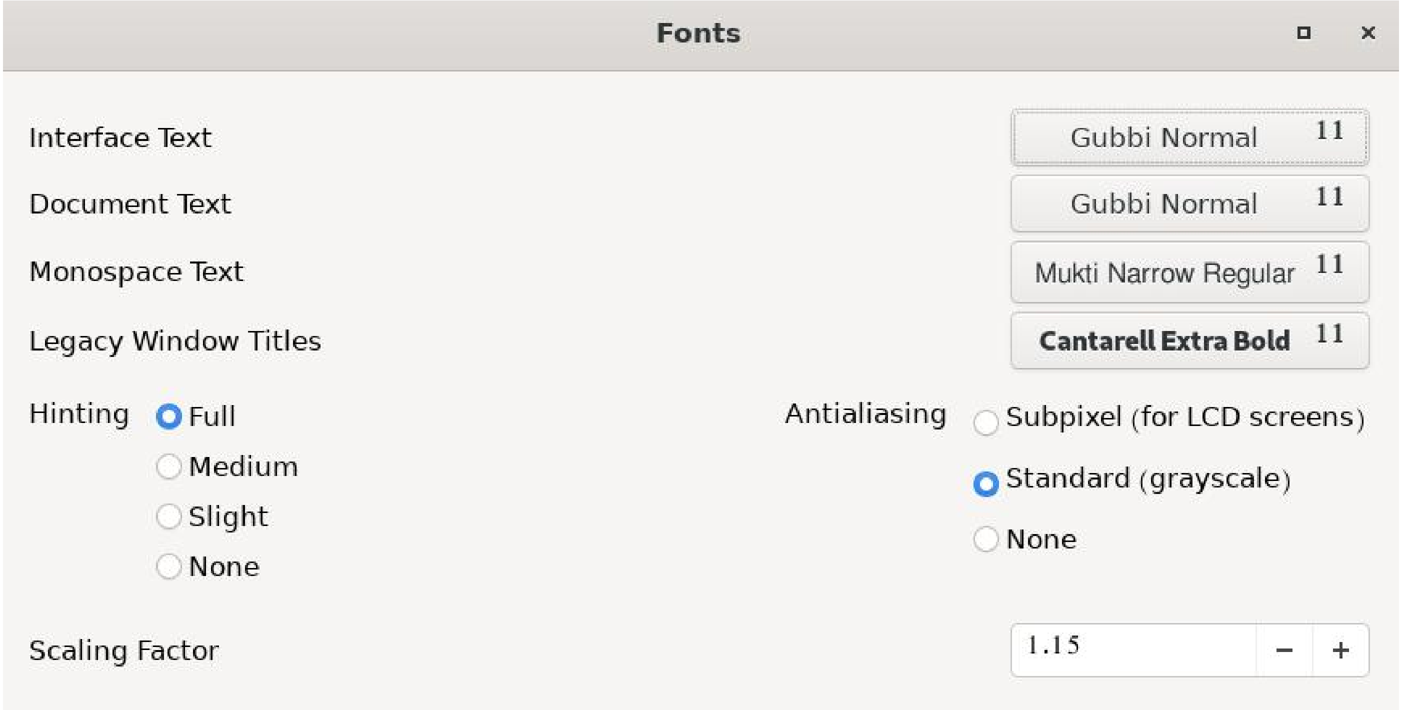

Lock Screen
Image
joshua-coleman-something-yellow.jpg
Adjustment
Zoom

3.4.4)


Icons
Adwaita (default)


Icons
DMZ-Black


Fonts
Interface Text
Cantarell Regular 11
Document Text
Cantarell Regular 11
Monospace Text
Monospace Regular 11
Legacy Window Titles
Cantarell Bold 11
Hinting
Full
Medium
Slight
None
Antialiasing
Subpixel (for LCD screens)
Standard (grayscale)
None
Scaling Factor
1.00


Fonts
Interface Text
Gubbi Normal 11
Document Text
Gubbi Normal 11
Monospace Text
Mukti Narrow Regular 11
Legacy Window Titles
Cantarell Extra Bold 11
Hinting
Full
Medium
Slight
None
Antialiasing
Subpixel (for LCD screens)
Standard (grayscale)
None
Scaling Factor
1.15
−
+

Bibliography

Digital Djinn, 2023. *A Comprehensive Exploration of the Advantages and Disadvantages of Kali Linux OS.* [Online]
Available at: https://medium.com/@digitaldjinn.official/a-comprehensive-exploration-of-the-advantages-and-disadvantages-of-kali-linux-os-0598ba0bd32f
[Accessed 2 June 2024].

Kali Linux, 2024. *What is Kali Linux?.* [Online]
Available at: https://www.kali.org/docs/introduction/what-is-kali-linux/
[Accessed 2 June 2024].

Nonis, S., 2024. *What is Ubuntu? Ubuntu Features, Advantages, and Disadvantages.* [Online]
Available at: https://monovm.com/blog/what-is-ubuntu/
[Accessed 2 June 2024].

Prakash, A., 2023. *How to Use GNOME Shell Extensions [Complete Guide].* [Online]
Available at: https://itsfoss.com/gnome-shell-extensions/
[Accessed 19 June 2024].

Sajee, S., 2023. *Pros and cons of Ubuntu OS.* [Online]
Available at: https://medium.com/@sajsajee18/pros-and-cons-of-ubuntu-os-667a5562137a
[Accessed 2 June 2024].

Ubuntu, 2024. *The story of Ubuntu.* [Online]
Available at: https://ubuntu.com/about
[Accessed 2 June 2024].